QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

cairo @ 68a61f6c6251 · StarkWare — public repository (passive review)

Organization	QSI
Report date	2026-10-28 12:22 UTC
Report version	1
Audit run	3ee51131-9f40-4430-8d6c-442a9bdfd05e · Document audit · 2026-10-28 12:22 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Web3 project
Content SHA-256	`18d87b68c3d7d5ee5324566405f8f401703b54def1529b72cf769a0cbf89aa4a`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

24	1	4	5	10
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	19
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	2
Improvement recommendation	2
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-04420	Critical	zip 0.6.6 is affected by RUSTSEC-2026-0369	40.5
QSI-2026-04406	High	pulldown-cmark 0.13.4 is affected by RUSTSEC-2026-0414	49.2
QSI-2026-04408	High	quinn-proto 0.11.17 is affected by RUSTSEC-2026-0340	36.9
QSI-2026-04409	High	quinn-proto 0.11.17 is affected by RUSTSEC-2026-0343	36.9
QSI-2026-04418	High	zip 0.6.6 is affected by RUSTSEC-2026-0367	36.9

Mosca's inequality (x + y > z): x = 5, y = 5, z = 12 → tight margin. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/starkware-libs/cairo @ 68a61f6c6251900e0080e09c812ffaeb37fd358e — evidence links open the lines at this commit.

Supplied inputs

567 files, 8154078 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Web3 project" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
dependencies	checked	50	Cargo.lock: 453 dependencies parsed (453 pinned); Cargo.toml: 63 dependencies parsed (1 pinned); crates/bin/cairo-compile/Cargo.toml: 9 dependencies parsed (3 pinned)
source_code	checked	516	crates/bin/cairo-compile/src/main.rs: 86 lines scanned with 16 rule patterns and 122 registry patterns; crates/bin/cairo-execute/src/main.rs: 452 lines scanned with 16 rule patterns and 122 registry patterns; crates/bin/cairo-format/src/main.rs: 232 lines scanned with 16 rule patterns and 122 registry patterns
osv_advisories	checked	1	485 pinned packages queried at OSV.dev; 13 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 567 (input files: C)

Architecture overview

Static review of the source tree cairo at 68a61f6c6251900e0080e09c812ffaeb37fd358e. 567 files analysed; of 1762 files, 855 are test, fuzzing, benchmark or example code and were left out, 340 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. No requests were sent to the owner's systems.

Overall security posture

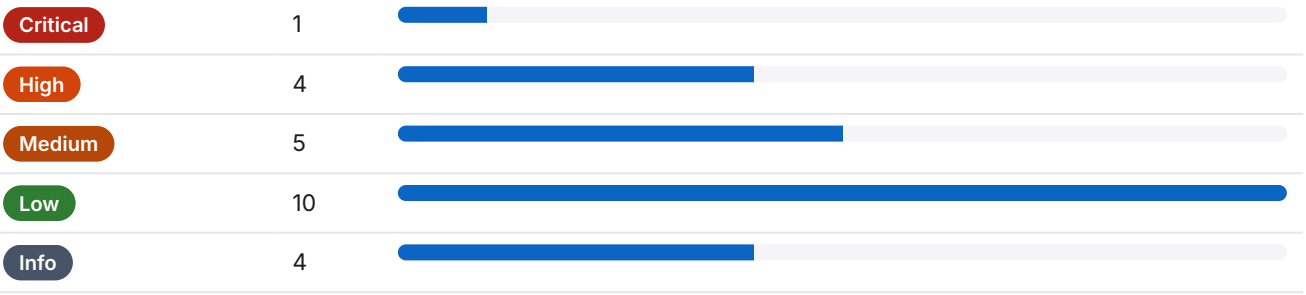
2
Indicative posture index (0–100)

55.6%
Quantum-vulnerable share of inventory

18
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-04420	zip 0.6.6 is affected by RUSTSEC-2026-0369	Known vulnerable component	Critical	Low
QSI-2026-04406	pulldown-cmark 0.13.4 is affected by RUSTSEC-2026-0414	Known vulnerable component	High	Medium
QSI-2026-04408	quinn-proto 0.11.17 is affected by RUSTSEC-2026-0340	Known vulnerable component	High	Low
QSI-2026-04409	quinn-proto 0.11.17 is affected by RUSTSEC-2026-0343	Known vulnerable component	High	Low
QSI-2026-04418	zip 0.6.6 is affected by RUSTSEC-2026-0367	Known vulnerable component	High	Low
QSI-2026-04398	Quantum-vulnerable ECDSA: transaction and message signing (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-04397	Quantum-vulnerable ECDSA: role not determined (4 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-04401	block-buffer 0.10.4 is affected by GHSA-qwgh-2vcv-g2f7	Known vulnerable component	Medium	Low
QSI-2026-04410	rustls 0.23.43 is affected by GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285)	Known vulnerable component	Medium	Low
QSI-2026-04411	rustls 0.23.43 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)	Known vulnerable component	Medium	Low
QSI-2026-04404	lru 0.12.5 is affected by GHSA-rhfx-m35p-ff5j (RUSTSEC-2026-0002)	Known vulnerable component	Low	Low
QSI-2026-04413	rustls 0.23.43 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)	Known vulnerable component	Low	Low
QSI-2026-04412	rustls 0.23.43 is affected by GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04402	crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04405	lru 0.12.5 is affected by RUSTSEC-2026-0253 (no severity rating)	Known vulnerable component	Low	Low, preliminary

QSI-2026-04407	quinn-proto 0.11.17 is affected by RUSTSEC-2026-0338 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04414	rustls-webpki 0.103.14 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04415	salsa 0.28.2 is affected by RUSTSEC-2026-0308 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04417	webpki-root-certs 1.0.9 is affected by RUSTSEC-2026-0383 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04419	zip 0.6.6 is affected by RUSTSEC-2026-0368 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-04403	derivative 2.2.0: informational advisory RUSTSEC-2024-0388 (unmaintained)	Improvement recommendation	Info	High
QSI-2026-04416	tokio 1.53.1 is affected by RUSTSEC-2026-0386 (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-04399	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium, preliminary
QSI-2026-04400	atomic-polyfill 1.0.3: informational advisory RUSTSEC-2023-0089 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-04420 · DEP-KNOWN-ADVISORY

Critical

zip 0.6.6 is affected by RUSTSEC-2026-0369

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0369 for crates.io package zip at version 0.6.6: ZipStreamReader::extract() and ZipArchive::extract() apply Unix permissions through symlinks without confining the target. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/zip@0.6.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	40.5 / 100 (Web3 project)
CVSS v3.1	9.1 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0369
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0369 · manifest · SHA-256 889fd754651c2317…


```
{"id": "RUSTSEC-2026-0369", "aliases": ["GHSA-4684-h2q9-2vcp"], "modified":
"2026-10-16T23:30:02.387698467Z", "summary": "ZipStreamReader::extract() and ZipArchive::extract()
apply Unix permissions through symlinks without confining the target", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (9.0.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0369

QSI-2026-04406 · DEP-KNOWN-ADVISORY

High

pulldown-cmark 0.13.4 is affected by RUSTSEC-2026-0414

Known vulnerable component | Confidence: Medium

OSV.dev reports RUSTSEC-2026-0414 for crates.io package pulldown-cmark at version 0.13.4: Exponential event growth with `ENABLE_WIKILINKS`. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/pulldown-cmark@0.13.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	49.2 / 100 (Web3 project)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0414
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0414 · manifest · SHA-256 84141280a2233722...


```
{"id": "RUSTSEC-2026-0414", "aliases": [], "modified": "2026-10-28T09:00:02.341043798Z", "summary":
"Exponential event growth with `ENABLE_WIKILINKS`", "severity": [{"type": "CVSS_V3", "score":
"CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.13.5).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0414

QSI-2026-04408 · DEP-KNOWN-ADVISORY

High

quinn-proto 0.11.17 is affected by RUSTSEC-2026-0340

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0340 for crates.io package quinn-proto at version 0.11.17: Peer minimum ACK delay can panic quinn-proto when ACK_FREQUENCY transmission is enabled. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.17
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	36.9 / 100 (Web3 project)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0340
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0340 · manifest · SHA-256 20321fd13a708b93...


```
{"id": "RUSTSEC-2026-0340", "aliases": ["GHSA-3g6h-r5qx-f6hq"], "modified":
"2026-10-12T18:15:01.975478256Z", "summary": "Peer minimum ACK delay can panic quinn-proto when
ACK_FREQUENCY transmission is enabled", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/
AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.11.18).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0340

QSI-2026-04409 · DEP-KNOWN-ADVISORY

High

quinn-proto 0.11.17 is affected by RUSTSEC-2026-0343

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0343 for crates.io package quinn-proto at version 0.11.17: Outgoing datagram buffer accounting can panic the sender in quinn-proto. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.17
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	36.9 / 100 (Web3 project)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0343
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0343 · manifest · SHA-256 8038e96bc82cf763...


```
{"id": "RUSTSEC-2026-0343", "aliases": ["GHSA-ppcp-v39w-8jq2"], "modified":
"2026-10-12T18:15:01.961745820Z", "summary": "Outgoing datagram buffer accounting can panic the
sender in quinn-proto", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/
C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.11.18).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0343

QSI-2026-04418 · DEP-KNOWN-ADVISORY

High

zip 0.6.6 is affected by RUSTSEC-2026-0367

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0367 for crates.io package zip at version 0.6.6: Insecure Preservation of SUID/SGID Bits during ZIP Extraction Leading to Potential Privilege Escalation. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/zip@0.6.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	36.9 / 100 (Web3 project)
CVSS v3.1	7.3 — CVSS:3.1/AV:L/AC:L/PR:L/UI:R/S:U/C:H/I:H/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0367
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0367 · manifest · SHA-256 db37f0f24f5796f6...


```
{"id": "RUSTSEC-2026-0367", "aliases": ["GHSA-vv5m-jxxh-4627"], "modified":
"2026-10-16T23:30:02.392172556Z", "summary": " Insecure Preservation of SUID/SGID Bits during ZIP
Extraction Leading to Potential Privilege Escalation", "severity": [{"type": "CVSS_V3", "score":
"CVSS:3.1/AV:L/AC:L/PR:L/UI:R/S:U/C:H/I:H/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (9.0.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0367

QSI-2026-04398 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to ECDSA across 2 file(s), e.g. crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs:24. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 12:22 UTC
Priority score	33.0 / 100 (Web3 project)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=12.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs:24 and inspect the quoted
lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs · line 24 · SHA-256 114a4f0ae1d81020… · File 74c57ba011fd1453…

```
23 | use secp256::{Secp256Libfunc, Secp256PointType};
24 | pub mod secp256k1;
25 | pub mod secp256r1;
```

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs · line 25 · SHA-256 7e6e9c2b04ea6d15… · File 74c57ba011fd1453…

```
24 | pub mod secp256k1;
25 | pub mod secp256r1;
26 |
```

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256k1.rs · line 24 · SHA-256 ae5dfb6aa4a275c5… · File f70b17f0e1a3712c…

```
23 |     const TYPE_ID: GenericTypeId = Secp256k1PointType::ID;
24 |     const TYPE_ID_STR: &'static str = "core::starknet::secp256k1::Secp256k1Point";
25 | }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-04397 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (4 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

7 reference(s) to ECDSA across 4 file(s), e.g. crates/cairo-lang-runner/src/casm_run/mod.rs:41. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	crates/cairo-lang-runner/src/casm_run/mod.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 12:22 UTC
Priority score	31.7 / 100 (Web3 project)
CVSS v3.1	not applicable

Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=12.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/cairo-lang-runner/src/casm_run/mod.rs:41 and inspect the quoted lines; the evidence
lists further files.
```

EVIDENCE

file_excerpt · crates/cairo-lang-runner/src/casm_run/mod.rs · line 41 · SHA-256 0e71c46f74a3ab05… · File 95c92cb2bb92a68f…

```
   40 | use lambdaworks_math::cyclic_group::IsGroup;
   41 | use
lambdaworks_math::elliptic_curve::short_weierstrass::curves::secp256k1::curve::Secp256k1Curve;
   42 | use
lambdaworks_math::elliptic_curve::short_weierstrass::curves::secp256r1::curve::Secp256r1Curve;
```

file_excerpt · crates/cairo-lang-runner/src/casm_run/mod.rs · line 42 · SHA-256 f4453e699fd83272… · File 95c92cb2bb92a68f…

```
   41 | use
lambdaworks_math::elliptic_curve::short_weierstrass::curves::secp256k1::curve::Secp256k1Curve;
   42 | use
lambdaworks_math::elliptic_curve::short_weierstrass::curves::secp256r1::curve::Secp256r1Curve;
   43 | use
lambdaworks_math::elliptic_curve::short_weierstrass::point::ShortWeierstrassProjectivePoint;
```

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs · line 3 · SHA-256 2c337fb5211b6a33… · File f82f3b08cb95fbe1…

```
    2 |
    3 | use super::secp256k1::{Secp256k1, Secp256k1PointType};
    4 | use super::secp256r1::{Secp256r1, Secp256r1PointType};
```

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs · line 4 · SHA-256 20087e0fcacf6d88… · File f82f3b08cb95fbe1…

```
    3 | use super::secp256k1::{Secp256k1, Secp256k1PointType};
    4 | use super::secp256r1::{Secp256r1, Secp256r1PointType};
    5 | use super::syscalls::SyscallGenericLibfunc;
```

file_excerpt · crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256r1.rs · line 24 · SHA-256 d4163de088e926b5… · File ea55c773bb58e7ac…

```
   23 |     const TYPE_ID: GenericTypeId = Secp256r1PointType::ID;
   24 |     const TYPE_ID_STR: &'static str = "core::starknet::secp256r1::Secp256r1Point";
   25 | }
```

file_excerpt · crates/cairo-lang-starknet-classes/src/felt252_serde.rs · line 11 · SHA-256 1eb81a2d0feea5c6… · File 4d25743988ab34f0…

```
   10 | use cairo_lang_sierra::extensions::starknet::secp256::Secp256GetPointFromXLibfunc;
   11 | use cairo_lang_sierra::extensions::starknet::secp256k1::Secp256k1;
   12 | use cairo_lang_sierra::extensions::starknet::secp256r1::Secp256r1;
```

file_excerpt · crates/cairo-lang-starknet-classes/src/felt252_serde.rs · line 12 · SHA-256 f63ec86cc34d6c40… · File 4d25743988ab34f0…

```
11 | use cairo_lang_sierra::extensions::starknet::secp256k1::Secp256k1;
12 | use cairo_lang_sierra::extensions::starknet::secp256r1::Secp256r1;
13 | use cairo_lang_sierra::extensions::starknet::storage::{
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-04401 · DEP-KNOWN-ADVISORY

Medium

block-buffer 0.10.4 is affected by GHSA-qwgh-2vcv-g2f7

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-qwgh-2vcv-g2f7 for crates.io package block-buffer at version 0.10.4: block_buffer: panic corrupts inline buffer position. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/block-buffer@0.10.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	31.5 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-qwgh-2vcv-g2f7
```

EVIDENCE

tool_output · OSV.dev GHSA-qwgh-2vcv-g2f7 · manifest · SHA-256 ac22c9ee6301d8c1...


```
{"id": "GHSA-qwgh-2vcv-g2f7", "aliases": [], "modified": "2026-08-19T19:30:06.686866977Z",
"summary": "block_buffer: panic corrupts inline buffer position", "severity": [{"type": "CVSS_V4",
"score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/VC:L/VI:L/VA:N/SC:N/SI:N/SA:N"}],
"declared_in_manifest": false}
```

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (none published yet; see the advisory for mitigations).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-qwgh-2vcv-g2f7

QSI-2026-04410 · DEP-KNOWN-ADVISORY

Medium

rustls 0.23.43 is affected by GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285) for crates.io package rustls at version 0.23.43: Rustls: TLS 1.3 handshake messages incorrectly accepted across encryption level boundaries. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.43
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	31.5 / 100 (Web3 project)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-2mjx-qc3c-rqvc
```

EVIDENCE

tool_output · OSV.dev GHSA-2mjx-qc3c-rqvc · manifest · SHA-256 f20e4b21c01d3d75...


```
{"id": "GHSA-2mjx-qc3c-rqvc", "aliases": ["RUSTSEC-2026-0285"], "modified":
"2026-10-27T15:30:05.509305490Z", "summary": "Rustls: TLS 1.3 handshake messages incorrectly
accepted across encryption level boundaries", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/
AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.45).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-2mjx-qc3c-rqvc

QSI-2026-04411 · DEP-KNOWN-ADVISORY

Medium

rustls 0.23.43 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346) for crates.io package rustls at version 0.23.43: Rustls: Session Debug formatting exposes TLS master secrets and resumption PSKs. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.43
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	31.5 / 100 (Web3 project)
CVSS v3.1	4.0 — CVSS:3.1/AV:L/AC:H/PR:H/UI:R/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-c8m5-c952-6cm4
```

EVIDENCE

tool_output · OSV.dev GHSA-c8m5-c952-6cm4 · manifest · SHA-256 87554a0cce34470a...


```
{"id": "GHSA-c8m5-c952-6cm4", "aliases": ["RUSTSEC-2026-0346"], "modified":
"2026-10-27T16:15:05.354395696Z", "summary": "Rustls: Session Debug formatting exposes TLS master
secrets and resumption PSKs", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:L/AC:H/PR:H/
UI:R/S:U/C:H/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-c8m5-c952-6cm4

QSI-2026-04404 · DEP-KNOWN-ADVISORY

Low

lru 0.12.5 is affected by GHSA-rhfx-m35p-ff5j (RUSTSEC-2026-0002)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-rhfx-m35p-ff5j (RUSTSEC-2026-0002) for crates.io package lru at version 0.12.5: `IterMut` violates Stacked Borrows by invalidating internal pointer. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/lru@0.12.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	27.0 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-rhfx-m35p-ff5j
```

EVIDENCE

tool_output · OSV.dev GHSA-rhfx-m35p-ff5j · manifest · SHA-256 74c3e26941575e81…


```
{"id": "GHSA-rhfx-m35p-ff5j", "aliases": ["RUSTSEC-2026-0002"], "modified":
"2026-09-10T03:50:59.064354050Z", "summary": "`IterMut` violates Stacked Borrows by invalidating
internal pointer", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/
VC:N/VI:L/VA:N/SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.16.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-rhfx-m35p-ff5j

QSI-2026-04413 · DEP-KNOWN-ADVISORY

Low

rustls 0.23.43 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349) for crates.io package rustls at version 0.23.43: rustls: An unauthenticated ServerHello poisons the per-name key-exchange

record, so later ClientHellos carry only a classical X25519 share. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.43
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	27.0 / 100 (Web3 project)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-w44r-93j7-qv28
```

EVIDENCE

tool_output · OSV.dev GHSA-w44r-93j7-qv28 · manifest · SHA-256 f0fcaa2fa7a56e18...


```
{"id": "GHSA-w44r-93j7-qv28", "aliases": ["RUSTSEC-2026-0349"], "modified":
"2026-10-27T16:15:05.340347056Z", "summary": "rustls: An unauthenticated ServerHello poisons the
per-name key-exchange record, so later ClientHellos carry only a classical X25519 share",
"severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N"}],
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-w44r-93j7-qv28

QSI-2026-04412 · DEP-KNOWN-ADVISORY

Low

rustls 0.23.43 is affected by GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348) for crates.io package rustls at version 0.23.43: Rustls: Brotli certificate decompression can allocate an attacker-controlled amount of memory up to 1 GiB before authentication. The affected code sits behind the `brotli` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/rustls@0.23.43
Data source	OSV.dev

Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	22.9 / 100 (Web3 project)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-h2v8-394m-p7vv
```

EVIDENCE

tool_output · OSV.dev GHSA-h2v8-394m-p7vv · manifest · SHA-256 9118726d52fea7ce...


```
{"id": "GHSA-h2v8-394m-p7vv", "aliases": ["RUSTSEC-2026-0348"], "modified":
"2026-10-27T16:15:05.337819489Z", "summary": "Rustls: Brotli certificate decompression can allocate
an attacker-controlled amount of memory up to 1 GiB before authentication.", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-h2v8-394m-p7vv

QSI-2026-04402 · DEP-ADVISORY-UNRATED

Low

crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.5: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95...


```
{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary":
"Multiple timing variability issues in operations intended to be constant-time", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-04405 · DEP-ADVISORY-UNRATED

Low

lru 0.12.5 is affected by RUSTSEC-2026-0253 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0253 for crates.io package lru at version 0.12.5: Potential use-after-free due to lack of panic safety in `LruCache::pop()`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/lru@0.12.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0253
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0253 · manifest · SHA-256 06df98c535068998…


```
{"id": "RUSTSEC-2026-0253", "aliases": [], "modified": "2026-08-11T11:15:06.101553049Z", "summary":
"Potential use-after-free due to lack of panic safety in `LruCache::pop()`", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.18.2) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0253

QSI-2026-04407 · DEP-ADVISORY-UNRATED

Low

quinn-proto 0.11.17 is affected by RUSTSEC-2026-0338 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0338 for crates.io package quinn-proto at version 0.11.17: Remote peer can panic quinn-proto with zero-length STREAM frames during unordered reads. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.17
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0338
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0338 · manifest · SHA-256 2d8a96c1ccb5782d…

{"id": "RUSTSEC-2026-0338", "aliases": ["GHSA-465w-v9q3-7j98"], "modified": "2026-10-12T18:15:01.969514645Z", "summary": "Remote peer can panic quinn-proto with zero-length STREAM frames during unordered reads", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.11.18) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0338

QSI-2026-04414 · DEP-ADVISORY-UNRATED

Low

rustls-webpki 0.103.14 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0382 (CVE-2026-111943) for crates.io package rustls-webpki at version 0.103.14: Certificate verification may accept distrusted intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.14
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0382
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0382 · manifest · SHA-256 9111e73d75239253…

{"id": "RUSTSEC-2026-0382", "aliases": ["CVE-2026-111943", "GO-2026-6809", "RUSTSEC-2026-0381", "RUSTSEC-2026-0383"], "modified": "2026-10-21T00:55:56.763143468Z", "summary": "Certificate verification may accept distrusted intermediates", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0382

QSI-2026-04415 · DEP-ADVISORY-UNRATED

Low

salsa 0.28.2 is affected by RUSTSEC-2026-0308 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0308 for crates.io package salsa at version 0.28.2: Use-after-free in interned values and cached function results. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/salsa@0.28.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0308
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0308 · manifest · SHA-256 580ef2afe3fb4006…


```
{"id": "RUSTSEC-2026-0308", "aliases": ["GHSA-xc3w-55vh-cw3w"], "modified":
"2026-09-24T14:45:04.873565398Z", "summary": "Use-after-free in interned values and cached function
results", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.28.5) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0308

QSI-2026-04417 · DEP-ADVISORY-UNRATED

Low

webpki-root-certs 1.0.9 is affected by RUSTSEC-2026-0383 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0383 (CVE-2026-111943) for crates.io package webpki-root-certs at version 1.0.9: Bundled root certificates allow trust in distrusted emSign intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/webpki-root-certs@1.0.9
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0383
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0383 · manifest · SHA-256 5020317ed72c7e0c...


```
{"id": "RUSTSEC-2026-0383", "aliases": ["CVE-2026-111943", "GO-2026-6809", "RUSTSEC-2026-0381",
"RUSTSEC-2026-0382"], "modified": "2026-10-21T06:30:02.691648061Z", "summary": "Bundled root
certificates allow trust in distrusted emSign intermediates", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.26.11, 1.0.10) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0383

QSI-2026-04419 · DEP-ADVISORY-UNRATED

Low

zip 0.6.6 is affected by RUSTSEC-2026-0368 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0368 for crates.io package zip at version 0.6.6: zip: `ZipArchive::extract` creates symbolic links whose attacker-controlled targets are outside the destination directory. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/zip@0.6.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.4 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0368
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0368 · manifest · SHA-256 77cc28045dc809d1…


```
{"id": "RUSTSEC-2026-0368", "aliases": ["GHSA-4v9h-p7m9-rh26"], "modified":
"2026-10-16T23:30:02.383750571Z", "summary": "zip: `ZipArchive::extract` creates symbolic links
whose attacker-controlled targets are outside the destination directory", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (9.0.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0368

QSI-2026-04403 · DEP-ADVISORY-INFORMATIONAL

Info

derivative 2.2.0: informational advisory RUSTSEC-2024-0388 (unmaintained)

Improvement recommendation | Confidence: High

OSV.dev lists RUSTSEC-2024-0388 for crates.io package derivative at version 2.2.0 as informational (unmaintained): `derivative` is unmaintained; consider using an alternative. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/derivative@2.2.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	18.7 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2024-0388
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2024-0388 · manifest · SHA-256 0624a2acbe96d2e8...


```
{"id": "RUSTSEC-2024-0388", "aliases": [], "modified": "2024-11-10T13:34:40Z", "summary":
"`derivative` is unmaintained; consider using an alternative", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2024-0388

QSI-2026-04416 · DEP-ADVISORY-UNRATED

Info

tokio 1.53.1 is affected by RUSTSEC-2026-0386 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0386 for crates.io package tokio at version 1.53.1: `AsyncFd::new` and related constructors violate I/O safety. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (tokio::io::unix::AsyncFd::new, tokio::io::unix::AsyncFd::try_new, tokio::io::unix::AsyncFd::try_with_interest, tokio::io::unix::AsyncFd::with_interest) is named in the

audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:crates.io/tokio@1.53.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	15.3 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0386
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0386 · manifest · SHA-256 1ebb594fd9923dc0...


```
{"id": "RUSTSEC-2026-0386", "aliases": [], "modified": "2026-10-23T14:45:03.847864944Z", "summary":
"`AsyncFd::new` and related constructors violate I/O safety", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.51.6, 1.53.3) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0386

QSI-2026-04399 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium | preliminary — requires further verification

Across 50 dependency manifests, QSI found crypto libraries (openssl, ring, rustls) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	12.2 / 100 (Web3 project)

CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · Cargo.lock, Cargo.toml, crates/bin/cairo-compile/Cargo.toml, crates/bin/cairo-execute/Cargo.toml, crates/bin/cairo-format/Cargo.toml, crates/bin/cairo-run/Cargo.toml, crates/bin/cairo-size-profiler/Cargo.toml, crates/bin/generate-syntax/Cargo.toml, crates/bin/get-lowering/Cargo.toml, crates/bin/sierra-compile/Cargo.toml, crates/bin/starknet-compile/Cargo.toml, crates/bin/starknet-sierra-compile/Cargo.toml, crates/bin/starknet-sierra-extract-code/Cargo.toml, crates/bin/starknet-sierra-upgrade-val · SHA-256 1980ed6357825cc4…

```
Crypto libraries: openssl, ring, rustls. PQC support depends on the version for: openssl
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

QSI-2026-04400 · DEP-ADVISORY-INFORMATIONAL

Info

atomic-polyfill 1.0.3: informational advisory RUSTSEC-2023-0089 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2023-0089 for crates.io package atomic-polyfill at version 1.0.3 as informational (unmaintained): atomic-polyfill is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/atomic-polyfill@1.0.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 12:22 UTC
Priority score	11.2 / 100 (Web3 project)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2023-0089
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2023-0089 · manifest · SHA-256 0fcfb6f26f61ecb9...


```
{"id": "RUSTSEC-2023-0089", "aliases": [], "modified": "2025-04-29T08:31:10Z", "summary": "atomic-polyfill is unmaintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2023-0089

Evidence

ID	Type	Data source	SHA-256
QSI-2026-04420	tool_output	OSV.dev RUSTSEC-2026-0369 (manifest)	889fd754651c2317ee1367e6b79afa5c8dc85363614c3ddf017e6e7118c94
QSI-2026-04406	tool_output	OSV.dev RUSTSEC-2026-0414 (manifest)	84141280a223372225a976433f4a051a0f151b4a2d3daaa5d454b3808029
QSI-2026-04408	tool_output	OSV.dev RUSTSEC-2026-0340 (manifest)	20321fd13a708b936ab6ccc115a6ed97f9b78eb43a4e76d134ec1664a5fc8
QSI-2026-04409	tool_output	OSV.dev RUSTSEC-2026-0343 (manifest)	8038e96bc82cf763be522e034a1baf7a01bd0b6523ae60faa215d96e1fd3a
QSI-2026-04418	tool_output	OSV.dev RUSTSEC-2026-0367 (manifest)	db37f0f24f5796f64e8d2e96846b7cf21dd61e1b0a4c7de21b8268aa1a29a
QSI-2026-04398	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs (line 24)	114a4f0ae1d810207edaf43599b7cc9c2dd4f7800eb06b14137002a53c4d1
QSI-2026-04398	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs (line 25)	7e6e9c2b04ea6d156352d55791e4b83d960da30ac66655bad69061284c310
QSI-2026-04398	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256k1.rs (line 24)	ae5dfb6aa4a275c5330cd1255673964bbd5607c4d8f7d91a489d41672690c
QSI-2026-04397	file_excerpt	crates/cairo-lang-runner/src/casm_run/mod.rs (line 41)	0e71c46f74a3ab0591cba43edd9de9663e8a101e5386eebba48bc383848c9
QSI-2026-04397	file_excerpt	crates/cairo-lang-runner/src/casm_run/mod.rs (line 42)	f4453e699fd83272c62133e0dd37b99331fbdec4356805860f334ca4b867a
QSI-2026-04397	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs (line 3)	2c337fb5211b6a3336031a7e331caf1e52a0f93519608e9f0dd756fc0e397
QSI-2026-04397	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs (line 4)	20087e0fcacf6d88cdd30d21dd6df4607503072361aee4392c24edb9a627

QSI-2026-04397	file_excerpt	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256r1.rs (line 24)	d4163de088e926b53c4c5dce2c77bb74aeba87048a8ec413c75a07e5fd07
QSI-2026-04397	file_excerpt	crates/cairo-lang-starknet-classes/src/felt252_serde.rs (line 11)	1eb81a2d0feea5c6487b656df37086af50f29a08d0020b120d765748e7a7
QSI-2026-04397	file_excerpt	crates/cairo-lang-starknet-classes/src/felt252_serde.rs (line 12)	f63ec86cc34d6c40745db6eb5255f177f151fe4b72c607f13e481d4acd9c
QSI-2026-04401	tool_output	OSV.dev GHSA-qwgh-2vcv-g2f7 (manifest)	ac22c9ee6301d8c1e4b09ca6951303d71527ccba2df30c96af92531d8efb
QSI-2026-04410	tool_output	OSV.dev GHSA-2mjx-qc3c-rqvc (manifest)	f20e4b21c01d3d75460ac8d90a54d541a1a34187a4f05b501e4f46c27c56
QSI-2026-04411	tool_output	OSV.dev GHSA-c8m5-c952-6cm4 (manifest)	87554a0cce34470a221f04ad57566b2b512ff6299ad3ee95b190df659d64
QSI-2026-04404	tool_output	OSV.dev GHSA-rhfx-m35p-ff5j (manifest)	74c3e26941575e818cf192f0511138a6719b2124292f8348b2d299a63109
QSI-2026-04413	tool_output	OSV.dev GHSA-w44r-93j7-qv28 (manifest)	f0fcaa2fa7a56e188458f39e55b23e999d0c7da6d3c1294c014c64ab37b3
QSI-2026-04412	tool_output	OSV.dev GHSA-h2v8-394m-p7vv (manifest)	9118726d52fea7cee09a6b4a21a10d2a7bab860f9762fff4658cb78db045
QSI-2026-04402	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567dfa2ba2667b
QSI-2026-04405	tool_output	OSV.dev RUSTSEC-2026-0253 (manifest)	06df98c5350689983b2fda96297e3527fd26f472c5d474c041975b220e7f
QSI-2026-04407	tool_output	OSV.dev RUSTSEC-2026-0338 (manifest)	2d8a96c1ccb5782dfd51ab6c6bfc548092df0a416eb5a7c29ca5b5ffdce6
QSI-2026-04414	tool_output	OSV.dev RUSTSEC-2026-0382 (manifest)	9111e73d752392534752d5ea5c9da4ec9a422a29c088531d446d16b27a01
QSI-2026-04415	tool_output	OSV.dev RUSTSEC-2026-0308 (manifest)	580ef2afe3fb4006afc362c0b351c3c07e95f925e97e474f8909ed94a7cb

QSI-2026-04417	tool_output	OSV.dev RUSTSEC-2026-0383 (manifest)	5020317ed72c7e0cbf2d22efcc385ee146b42c336ffbeb3bf453e64cec1e9
QSI-2026-04419	tool_output	OSV.dev RUSTSEC-2026-0368 (manifest)	77cc28045dc809d16dfd7075061aa99e7aa21d64c6bf6d60981905302d10a
QSI-2026-04403	tool_output	OSV.dev RUSTSEC-2024-0388 (manifest)	0624a2acbe96d2e8c44b2e7c9e0253aa4d99e8e969cd062a4b077f6a3074
QSI-2026-04416	tool_output	OSV.dev RUSTSEC-2026-0386 (manifest)	1ebb594fd9923dc03209b41738d3f9cf8f324941e8844709473c643cd247e
QSI-2026-04399	dependency	Cargo.lock, Cargo.toml, crates/bin/cairo-compile/Cargo.toml, crates/bin/cairo-execute/Cargo.toml, crates/bin/cairo-format/Cargo.toml, crates/bin/cairo-run/Cargo.toml, crates/bin/cairo-size-profiler/Cargo.toml, crates/bin/generate-syntax/Cargo.toml, crates/bin/get-lowering/Cargo.toml, crates/bin/sierra-compile/Cargo.toml, crates/bin/starknet-compile/Cargo.toml, crates/bin/starknet-sierra-compile/Cargo.toml, crates/bin/starknet-sierra-extract-code/Cargo.toml, crates/bin/starknet-sierra-upgrade-val	1980ed6357825cc444a27bc1b98985b873318b59d5a5fa25de06c96d7c8e
QSI-2026-04400	tool_output	OSV.dev RUSTSEC-2023-0089 (manifest)	0fcfb6f26f61ecb982bc38b6d6f12db7c77d0fc40bf9353a089129368e765

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	5	10
Migration time y	5	5
Assumed years to CRQC z	12 (central)	12
x + y − z	-2 → tight margin	3 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Web3 project”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (2)	QSI-2026-04398	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
role not determined	ECDSA (4)	QSI-2026-04397	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
Grover: acceptable to NIST, below ECCG/ANSSI guidance	3

n/a	5
vulnerable (Shor)	10

Detailed findings: QSI-2026-04398, QSI-2026-04397

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDSA (10)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Keccak-256 (3)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Keccak-256 (3)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-04420 · P2 · 30 days

root-cause fix

Upgrade zip to a fixed version (9.0.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0369.

QSI-2026-04406 · P2 · 30 days

root-cause fix

Upgrade pulldown-cmark to a fixed version (0.13.5).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0414.

QSI-2026-04408 · P2 · 30 days

root-cause fix

Upgrade quinn-proto to a fixed version (0.11.18).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0340.

QSI-2026-04409 · P2 · 30 days

root-cause fix

Upgrade quinn-proto to a fixed version (0.11.18).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.

Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0343.

QSI-2026-04418 · P2 · 30 days root-cause fix

Upgrade zip to a fixed version (9.0.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0367.

QSI-2026-04401 · P2 · 30 days root-cause fix

Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-qwgh-2vcv-g2f7.

QSI-2026-04410 · P2 · 30 days root-cause fix

Upgrade rustls to a fixed version (0.23.45).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285).

QSI-2026-04411 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346).

QSI-2026-04404 · P2 · 30 days

root-cause fix

Upgrade lru to a fixed version (0.16.3).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-rhfx-m35p-ff5j (RUSTSEC-2026-0002).

QSI-2026-04413 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349).

QSI-2026-04412 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low

Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348).

QSI-2026-04402 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-04405 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0253; upgrade lru to a fixed version (0.18.2) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0253, or the triage is recorded.

QSI-2026-04407 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0338; upgrade quinn-proto to a fixed version (0.11.18) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports RUSTSEC-2026-0338, or the triage is recorded.

QSI-2026-04414 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0382 (CVE-2026-111943), or the triage is recorded.

QSI-2026-04415 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0308; upgrade salsa to a fixed version (0.28.5) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0308, or the triage is recorded.

QSI-2026-04417 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0383 (CVE-2026-111943); upgrade webpki-root-certs to a fixed version (0.26.11, 1.0.10) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0383 (CVE-2026-111943), or the triage is recorded.

QSI-2026-04419 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0368; upgrade zip to a fixed version (9.0.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0368, or the triage is recorded.

QSI-2026-04416 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0386, or the triage is recorded.

QSI-2026-04398 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-04397 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-04398 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-04397 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-04403 · P4 · 90 days root-cause fix

Replace derivative with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	derivative replaced, or the decision recorded.

QSI-2026-04400 · P4 · 90 days root-cause fix

Replace atomic-polyfill with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	atomic-polyfill replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-04420	Upgrade zip to a fixed version (9.0.0).	root-cause fix
P2	QSI-2026-04406	Upgrade pulldown-cmark to a fixed version (0.13.5).	root-cause fix
P2	QSI-2026-04408	Upgrade quinn-proto to a fixed version (0.11.18).	root-cause fix
P2	QSI-2026-04409	Upgrade quinn-proto to a fixed version (0.11.18).	root-cause fix
P2	QSI-2026-04418	Upgrade zip to a fixed version (9.0.0).	root-cause fix
P2	QSI-2026-04401	Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).	root-cause fix
P2	QSI-2026-04410	Upgrade rustls to a fixed version (0.23.45).	root-cause fix
P2	QSI-2026-04411	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-04404	Upgrade lru to a fixed version (0.16.3).	root-cause fix
P2	QSI-2026-04413	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-04412	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-04402	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix
P3	QSI-2026-04405	Triage RUSTSEC-2026-0253; upgrade lru to a fixed version (0.18.2) if it applies.	root-cause fix
P3	QSI-2026-04407	Triage RUSTSEC-2026-0338; upgrade quinn-proto to a fixed version (0.11.18) if it applies.	root-cause fix
P3	QSI-2026-04414	Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.	root-cause fix
P3	QSI-2026-04415	Triage RUSTSEC-2026-0308; upgrade salsa to a fixed version (0.28.5) if it applies.	root-cause fix
P3	QSI-2026-04417	Triage RUSTSEC-2026-0383 (CVE-2026-111943); upgrade webpki-root-certs to a fixed version (0.26.11, 1.0.10) if it applies.	root-cause fix

P3	QSI-2026-04419	Triage RUSTSEC-2026-0368; upgrade zip to a fixed version (9.0.0) if it applies.	root-cause fix
P3	QSI-2026-04416	Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-04398	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-04397	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-04398	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-04397	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-04403	Replace derivative with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-04400	Replace atomic-polyfill with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-04420	Re-run the dependency scan after upgrading.
QSI-2026-04406	Re-run the dependency scan after upgrading.
QSI-2026-04408	Re-run the dependency scan after upgrading.
QSI-2026-04409	Re-run the dependency scan after upgrading.
QSI-2026-04418	Re-run the dependency scan after upgrading.
QSI-2026-04398	Re-run the inventory after refactoring.
QSI-2026-04397	Re-run the inventory after refactoring.
QSI-2026-04401	Re-run the dependency scan after upgrading.
QSI-2026-04410	Re-run the dependency scan after upgrading.
QSI-2026-04411	Re-run the dependency scan after upgrading.
QSI-2026-04404	Re-run the dependency scan after upgrading.
QSI-2026-04413	Re-run the dependency scan after upgrading.
QSI-2026-04412	Re-run the dependency scan after upgrading.
QSI-2026-04402	Re-run the dependency scan after upgrading.
QSI-2026-04405	Re-run the dependency scan after upgrading.
QSI-2026-04407	Re-run the dependency scan after upgrading.
QSI-2026-04414	Re-run the dependency scan after upgrading.
QSI-2026-04415	Re-run the dependency scan after upgrading.
QSI-2026-04417	Re-run the dependency scan after upgrading.
QSI-2026-04419	Re-run the dependency scan after upgrading.
QSI-2026-04403	Re-run the dependency scan after the change.
QSI-2026-04416	Re-run the dependency scan after upgrading.
QSI-2026-04399	Re-scan after upgrades.
QSI-2026-04400	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

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Reviewed: 2026-10-13

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31. OSV advisory RUSTSEC-2026-0343 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0343 (modified 2026-10-12T18:15:01.961745820Z)
32. OSV advisory RUSTSEC-2026-0367 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0367 (modified 2026-10-16T23:30:02.392172556Z)
33. OSV advisory RUSTSEC-2026-0368 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0368 (modified 2026-10-16T23:30:02.383750571Z)
34. OSV advisory RUSTSEC-2026-0369 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0369 (modified 2026-10-16T23:30:02.387698467Z)
35. OSV advisory RUSTSEC-2026-0382 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0382 (modified 2026-10-21T00:55:56.763143468Z)
36. OSV advisory RUSTSEC-2026-0383 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0383 (modified 2026-10-21T06:30:02.691648061Z)
37. OSV advisory RUSTSEC-2026-0386 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0386 (modified 2026-10-23T14:45:03.847864944Z)
38. OSV advisory RUSTSEC-2026-0387 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0387 (modified 2026-10-24T03:45:02.365137418Z)
39. OSV advisory RUSTSEC-2026-0414 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0414 (modified 2026-10-28T09:00:02.341043798Z)
40. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
41. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)
42. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
Keccak-256	Keccak-256	crates/cairo-lang-semantic/src/types.rs:23	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
Keccak-256	Keccak-256	crates/cairo-lang-sierra/src/ids.rs:4	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
Keccak-256	Keccak-256	crates/cairo-lang-starknet-classes/src/keccak.rs:2	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
Crypto library openssl (OpenSSL)	0.10.81	Cargo.lock	n/a	acceptable
Crypto library ring (ring (Rust))	0.17.14	Cargo.lock	n/a	acceptable
Crypto library rustls (rustls)	0.23.43	Cargo.lock	n/a	acceptable
Crypto library openssl (OpenSSL)	0.10.73	Cargo.toml	n/a	acceptable
Crypto library openssl (OpenSSL)	workspace	crates/bin/starknet-sierra-upgrade-validate/Cargo.toml	n/a	acceptable
ECDSA	secp256k1	crates/cairo-lang-runner/src/casm_run/mod.rs:41	vulnerable (Shor)	acceptable
ECDSA	P-256	crates/cairo-lang-runner/src/casm_run/mod.rs:42	vulnerable (Shor)	acceptable
ECDSA	secp256k1	crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs:24	vulnerable (Shor)	acceptable
ECDSA	P-256	crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs:25	vulnerable (Shor)	acceptable
ECDSA	secp256k1	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs:3	vulnerable (Shor)	acceptable
ECDSA	P-256	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs:4	vulnerable (Shor)	acceptable
ECDSA	secp256k1	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256k1.rs:24	vulnerable (Shor)	acceptable

ECDSA	P-256	crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256r1.rs:24	vulnerable (Shor)	acceptable
ECDSA	secp256k1	crates/cairo-lang-starknet-classes/src/felt252_serde.rs:11	vulnerable (Shor)	acceptable
ECDSA	P-256	crates/cairo-lang-starknet-classes/src/felt252_serde.rs:12	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 18d87b68c3d7d5ee5324566405f8f401703b54def1529b72cf769a0cbf89aa4a

C. Input files

File	Size	SHA-256	Red
Cargo.lock	125405	afaf48f30a42c610b766e6dcb06e68203d45e43c2f339bb882e0c6a03c2895f3	0
Cargo.toml	5333	4df1342b05c0d893010469be852f60c02d3829234da4e047393f1a4a89c76a02	0
crates/bin/cairo-compile/Cargo.toml	745	824cbadd1ac84f537ec2daf506eb0d9f63385e33e051c26dd25190ed002eff7c	0
crates/bin/cairo-compile/src/main.rs	2696	030cf60d8a534d663a8ad398b94c636fd3dc7dddbd6205466a11858f604f0f3c	0
crates/bin/cairo-execute/Cargo.toml	1131	0b7a69d84c623c1ac8fec351caaaf6017c91ce06a10f51d3708af43e223d0346	0
crates/bin/cairo-execute/src/main.rs	18076	e219a06602c041ea9808603f6ab977ec8d8895fbf45c19c1f243e1edd4fb8782	0
crates/bin/cairo-format/Cargo.toml	625	4df498087b2446916e7acc7590a7e447785c26f8448b8283c0a33904b289fa8c	0
crates/bin/cairo-format/src/main.rs	7798	a57c37c2692016394a1fac7099d939af6bf40d382a62d262750978cebe720786	0
crates/bin/cairo-run/Cargo.toml	872	e57a13f381ca79ac62689fb9bf505f42d65d8d39985ac3c44f829250e192f4d2	0
crates/bin/cairo-run/src/main.rs	5709	682584923a2e7b6f0771600966da3ff4d777c58101a200665441aad59cb9ed71	0
crates/bin/cairo-size-profiler/Cargo.toml	1632	08cd0c5b2ba09157ce4dbce0d4ec7b7402c90a5e6900750cdeeba7fffb3b9caa	0
crates/bin/cairo-size-profiler/src/main.rs	10510	feb0d747a61fbea953b534dd837470676ef70b119d3fc1c3e3161a0127027688	0
crates/bin/generate-syntax/Cargo.toml	416	389238bb35848ac3dcc7a30674dc2392678f67a30a713d86f319eb37c21823b3	0
crates/bin/generate-syntax/src/main.rs	381	2beb19a7d0b697c788286ab8a660190de330f1ed9f195c1434fb4ef6ab0920c1	0
crates/bin/get-lowering/Cargo.toml	1310	ef9abc3b48e904666ecbeab399abdc333a8cb87b9253e64d886107f17d58c0f2	0
crates/bin/get-lowering/src/main.rs	12193	e0b74b87a0fb1d88d3da807a455afac6e223bba37fdc80b3bb3ebbe84accf5c2	0
crates/bin/sierra-compile/Cargo.toml	812	9f4acba2e68a2a2954656e8a1ed69da0944e25dbd9eac141c3faf8d5ddb623fc	0

crates/bin/sierra-compile/src/main.rs	1928	4bd96f8e455cd35580518555b93a212750a93497f97db3fb22b74896fdb01e4e	0
crates/bin/starknet-compile/Cargo.toml	720	52ec2ca0cfac91e682978cb181819244f8aac806e9725ca3c385ee24a0bffb73	0
crates/bin/starknet-compile/src/main.rs	2879	5674b13d82ad3318c72c7b24a4f8daa4e2e8559790c9c34198da0b56a45d4c69	0
crates/bin/starknet-sierra-compile/Cargo.toml	763	b245d1e68b405eaeb6dcb4b0f90e7facd069d8913a4a64b58b09d205c4c07039	0
crates/bin/starknet-sierra-compile/src/main.rs	3149	14b41282cdcb1c5001cc22d55e2aa33ed90180a995af09c44c8d8d8fa5f37f8d	0
crates/bin/starknet-sierra-extract-code/Cargo.toml	534	4fb45462968ab82d980b4cdd071e0143892cfd3ecfea04eb403131adec40a2c8	0
crates/bin/starknet-sierra-extract-code/src/main.rs	1255	b5965f0e1d93d9909213cc9bf8dae4106f5971bde672be9abbb4094cfec5f517	0
crates/bin/starknet-sierra-upgrade-validate/Cargo.toml	1043	e0fb965ff51f06756aa2469c4e0439ebbfa74e99c90edec8d211675dbbfca957	0
crates/bin/starknet-sierra-upgrade-validate/src/main.rs	24474	2de6e8071de90fbebda78e6911339ba98a1d4d3a73dbfe6550b880c44518a8f4	0
crates/cairo-lang-casm/Cargo.toml	1157	ea552c74e97678a83934c352d40d5f7fcc6d5c0fd57875e0059e49e322cde86c	0
crates/cairo-lang-casm/src/ap_change.rs	4402	22e27ef0a9837285073b7ef657118eac32d8f0e33bca9a3c7d6746c2b2e0cecb	0
crates/cairo-lang-casm/src/assembler.rs	11348	11b70ff281ebde844f6bc0ea0a4a20cd327aafe7980b10987a2639bf2d78aec9	0
crates/cairo-lang-casm/src/builder.rs	43291	760f1cd450f3581b9efa4e3b0d8302be6e987990b6b699a2006c7dabdadd62c6	0
crates/cairo-lang-casm/src/cell_expression.rs	5178	62eadf9d821f94a4237ef5c0de4deec5e14d06531090eaeb671ba8a74f33f22d	0
crates/cairo-lang-casm/src/encoder.rs	4308	3a847ebfe4e8915235a5fca1b07b22bd1a430ee23c2719b87bec1e597ae82d24	0
crates/cairo-lang-casm/src/hints/mod.rs	37691	b20a3bb3d29fdaf7388fae70ec6208af59e1f121bdd6c7fd8144b7638974a9b7	0
crates/cairo-lang-casm/src/inline.rs	13207	15ecd50d7b5017f36de351e134face2c34e4516a85d2d9c32cff34a312e50707	0
crates/cairo-lang-casm/src/instructions.rs	6975	20797f3124c1feb700e86affab285ae722ee587bc9a84c318a10aa633d486aa7	0
crates/cairo-lang-casm/src/lib.rs	360	82c4240add8e990d62cd458398f111331de26bc78be7d38bc0cf3904823ffcc5	0
crates/cairo-lang-casm/src/operand.rs	5392	b0ab13c05e4a3c55152dc171aa8c6c921fd4a0138c8ac4c1dc5efa64acfb3ec5	0
crates/cairo-lang-compiler/Cargo.toml	1352	210db0b4cacc6000381bb3ed13cdd2a36f0d6cef28eeb34e42070c1613b886fe	0

crates/cairo-lang-compiler/src/db.rs	8614	771bb1082ed2f9a2fe47149fc557c02e99cb9f5e023b33e8e805836cf57b7d00	0
crates/cairo-lang-compiler/src/diagnostics.rs	11142	63434f06c1ccd42694618825e105f3e341e33b4e7adaaf744515ca3e1cdf3b4e	0
crates/cairo-lang-compiler/src/lib.rs	16895	75a24dd98333f88a5f0968c50013f43d76c51292fbbe0c7d6d3d8f4c20308843	0
crates/cairo-lang-compiler/src/project.rs	6212	5f2cbd7bd3bf374e12d6ba0b1bdc4359b26940f769f2f7fe169a13f9f72e3b6b	0
crates/cairo-lang-debug/Cargo.toml	552	f179b3d54b6aed01a71b64d463c25e14d01fd69800beaa76c738caf2fa2e68d1	0
crates/cairo-lang-debug/src/debug.rs	8539	12f7b5efd57aad97277639ae83045d584ccb521d8bc64a37fe345dc08aa2128c	0
crates/cairo-lang-debug/src/lib.rs	134	6db717119bad90e961d428dbbf650470d6742d42f11c850907a0e878fa118ddb	0
crates/cairo-lang-defs/Cargo.toml	1231	62dc4a2baa7da766e30e4408e9a7e7234fbeb3b9abac4893141273001471f861	0
crates/cairo-lang-defs/src/cache/mod.rs	88814	e9b19f19fbffba645de83ee79949c52a863ea67dea14fa35015ce361baf23820	0
crates/cairo-lang-defs/src/db.rs	77856	dcda77a40d2de9381c0734cf867c1cf62bc7641da2c865cd7fd6bb46475978eb	0
crates/cairo-lang-defs/src/diagnostic_utils.rs	3231	6f25738e325116ed65aac3b3cef4467f1a5574a1a83a0438c024f2f5b2566aca	0
crates/cairo-lang-defs/src/ids.rs	57051	149574dd264f072bb6c3a75d2be69ac3c4cd7d304c9b4fbb622f835c3a22a3cb	0
crates/cairo-lang-defs/src/lib.rs	331	3a751df9731e19e9831cfca8fbce13736f158134ebbc8ca78cefcaf1b106776c	0
crates/cairo-lang-defs/src/patcher.rs	13336	41b671f2542bb802b0402485884f6edb18b60da8081412519127b6039bb07006	0
crates/cairo-lang-defs/src/plugin.rs	9361	f9ee397d4705da5eff87d4a5317599535d0835c9d6beec9ea148fba5de27ad17	0
crates/cairo-lang-defs/src/plugin_utils.rs	8562	7114978b10f5eb7fa29c04c27bed72eb3163c220564e5f2dc8b720e4fa757a87	0
crates/cairo-lang-defs/src/stable_order.rs	4454	22b1155e82f8b9a945a51930f8916a1ed68713f881d1e886e422d32c226edd65	0
crates/cairo-lang-diagnostics/Cargo.toml	812	60fdbfacb1903bea67d204675fbfedff4ba1c38ca6b4b1cadad84cfaf6c3e40c	0
crates/cairo-lang-diagnostics/src/diagnostics.rs	14320	04ce38d55b23cc16b209fba0f3d2329ec43abb71770f21e3ccc3bafccdc51948	0
crates/cairo-lang-diagnostics/src/error_code.rs	2342	ff3b0a2276aeea4c8ebed7194ed9ecd7380eff4be786e643941ab9e868023c03	0
crates/cairo-lang-diagnostics/src/lib.rs	493	f5ccf1a38ab4b5b45ced31bffef7beb9e52ff8148260e2a31bd5f3fee805e77c	0
crates/cairo-lang-doc/Cargo.toml	1292	2c6c33eb61fdac82273ccc975e443d120c6bf0f8e348fa95b0505d9aba791109	0
crates/cairo-lang-doc/src/db.rs	11610	d91930171b6254904fb8470e598174190ed149b617081c56606296c3f6ec2537	0

crates/cairo-lang-doc/src/documentable_formatter.rs	29966	225519d3aaba9fca3e13863c920ad92889dfc09967695b298f45c4c21ab9a7f4	0
crates/cairo-lang-doc/src/documentable_item.rs	2281	b5f008ac9dd2b77cc2240aa1e85a15e0c5fe184c3e9ee5c3a5f9f7bf423c6a99	0
crates/cairo-lang-doc/src/helpers.rs	26774	f8140b146a3c8cc3d62d8c3f8f38146e4d67d519ccccdfcc32f1d4f6585f83b2	0
crates/cairo-lang-doc/src/lib.rs	203	cf6fa016b5838f157783bb2835742595ff68104bc5385b3b40a027a5fd57e69b	0
crates/cairo-lang-doc/src/location_links.rs	6446	e28cc9bd0932e8fc0e1b06d7aa2d552bd4b317c3144d1e6b108e4bdcd5ac6aeb	0
crates/cairo-lang-doc/src/parser.rs	18328	5fd4580addf52c253243710f51db1882dee3198233b9fc41a6936733b1592c4c	0
crates/cairo-lang-doc/src/signature_data.rs	24761	59277adfe9e188f2d80fdab7a90a8507bc7f1a55de394b0117313531b4423545	0
crates/cairo-lang-doc/src/signature_errors.rs	1479	502b9ab7cdf3cd7dda501fd40e2410273f68bdf627fa1c4296962665c1c342e4	0
crates/cairo-lang-eq-solver/Cargo.toml	473	6497a4901045844f2596e6807dfebcf429bfa1dd357da105306d8cb33b5fffb0	0
crates/cairo-lang-eq-solver/src/expr.rs	2023	ee026898443d81881f9537402b1fc05ac4081e50497ee1fbae103a15a64280c6	0
crates/cairo-lang-eq-solver/src/lib.rs	4380	7f062f8b517c533bb1e665f6055948168f720856b88bf796ed2250ea78fcd1e8	0
crates/cairo-lang-executable/Cargo.toml	1772	d1531c4e88f054b937b9cb7e503af83125c65645691de6613d70f45ea4b51a50	0
crates/cairo-lang-executable/src/compile.rs	10178	04cbd281e177bb6a0abf6a0ad7f5967eddd40ba8646c507e7e5991b9360f5adc	0
crates/cairo-lang-executable/src/debug_info.rs	2341	cf9161042c88e723361772cfbaa765724d85c372eee3708286706c5ae4999405	0
crates/cairo-lang-executable/src/executable.rs	3255	f29282d4022b9bc221ec2977b0e5acf7326d79f8b082e667f00fa93d6605be4d	0
crates/cairo-lang-executable/src/lib.rs	78	f90f222945c23a9ed5cf6fa81e612514fe592b434b78bd6687354a7fd35f7be1	0
crates/cairo-lang-executable-plugin/Cargo.toml	1086	5658a87167a22cd18bc848be57f3c04ae851ec8a2cfb791fa93727f845575571	0
crates/cairo-lang-executable-plugin/src/lib.rs	9302	391107d0bca32107336a7988582cf84c70d9d580fc70548ea63d8774e343fe6c	0
crates/cairo-lang-execute-utils/Cargo.toml	643	ba24ec61910b15a0c6473ef3fd1d0498a277858ad3f2d85d3d79c15484c66232	0
crates/cairo-lang-execute-utils/src/lib.rs	2728	a3328710869fc4071cb96292eb220a134784af68e08c6e935ba182d2bb18f08b	0
crates/cairo-lang-filesystem/Cargo.toml	957	64c4de286eee1585a0677f5b4b5c71a3e523524a142a0f7cfd7ebc3fd9a97b7e	0

crates/cairo-lang-filesystem/src/cfg.rs	5922	5dfa893eca474663a9edce475cb420a4237c58ab9c11bea506282404903617ae	0
crates/cairo-lang-filesystem/src/db.rs	26180	e9560e660025d8519c749c8ca2e16a03b8382cde576fbaa8be8b3d428d3559db	0
crates/cairo-lang-filesystem/src/detect.rs	865	e0a29d2945ce78e3a663cc8c9d4d9a2c399ccf8706ccef896559db8b5e4a0c3c	0
crates/cairo-lang-filesystem/src/flag.rs	5722	041ee696f404c616b75a24bbf6ad7420bdcb58631b6e016d2adc6bfdf0a7a809	0
crates/cairo-lang-filesystem/src/ids.rs	19041	9803c7b29b7e9225479088c55212134263efd212965784bde9bf90167f7d8841	0
crates/cairo-lang-filesystem/src/lib.rs	162	c08f14e13a5556d9b910ac52e4512968a43dd89f48b08912936911c7e55d7eb2	0
crates/cairo-lang-filesystem/src/location_marks.rs	4360	503db8406bad9e467a22f2e56c1bc5a91be6102154bff43370603f04fc4e2b19	0
crates/cairo-lang-filesystem/src/span.rs	8540	a430cc45addeefa8d64a6aba8cc33456bff165f981106decce2cbe65cd2d2145	0
crates/cairo-lang-formatter/Cargo.toml	931	ba6979b9b6799a00aa16c4ad2af948e56431c7ce57512a7eac573e970714d608	0
crates/cairo-lang-formatter/src/cairo_formatter.rs	9087	540ac75b48a0e14a79173317e1d8065d5f664e22db0db9d788e3f9481f457723	0
crates/cairo-lang-formatter/src/formatter_impl.rs	66995	3400918b8a7acd3a297db9cede1e7d66c7f5b289369e0f87946dffd691b64663	0
crates/cairo-lang-formatter/src/lib.rs	6049	73c35422e63e80f7b215687cbea6b1b7f49cdad4ba282662af7416b9b6b855ea	0
crates/cairo-lang-formatter/src/node_properties.rs	50675	57e5af74d68de2d4baf795f1ab6ec1c380af8eafba426867f9b639e657602433	0
crates/cairo-lang-lowering/Cargo.toml	1736	107702ddb74605781441e73d27f4a2a44980a87edd1157542a21f8655a89fbb3	0
crates/cairo-lang-lowering/src/add_withdraw_gas/mod.rs	4841	d29fac5c4bd5a25ebdb37bf38c0e615cec0b9f7cb58a4a2f79384a90cf2811ba	0
crates/cairo-lang-lowering/src/analysis/backward.rs	8996	345420cb955a8fd475d95efd16a3efaf77e8da428abd03232a9c8dc72c21b219	0
crates/cairo-lang-lowering/src/analysis/core.rs	7631	36daa787393b670fc9ebb5c94b0bf7ed7ba8885160c99bb86edda607e11c12b7	0
crates/cairo-lang-lowering/src/analysis/def_site.rs	4250	76fde60eed2cac9f793f9fa5c3ad81207bfe490541722353bb10cc406022c0bd	0
crates/cairo-lang-lowering/src/analysis/dominator.rs	3090	9d7db76b93adf03691f0eb028426a88f4380d7f5afac59446b55a6fac867ae9d	0
crates/cairo-lang-lowering/src/analysis/equality_analysis.rs	43371	162c2d54105a66a8a26f5ba57766deaf020b553dc5c16f265c08fa1eb0f31fc7	0

crates/cairo-lang-lowering/src/analysis/forward.rs	5911	726dfa12e1e0fef5757efcd34bf289496eeaa1e8d96a465bf59db7063c6fbe83	0
crates/cairo-lang-lowering/src/analysis/mod.rs	3726	d5aea227fce022d21414051dbcda16e1a7daf85dfad0a968b0b87e48c47c01bb	0
crates/cairo-lang-lowering/src/analysis/topological_order.rs	1883	c8d0ed6a85140d261dd616101c79f36ab1bdd975f41daea9ba583e1a2ea87c4a	0
crates/cairo-lang-lowering/src/analysis/use_sites.rs	4194	a864e2433bcba8b874b1917a2ce19c09c4a55aec5c9e6bf66701d0c283005f4e	0
crates/cairo-lang-lowering/src/borrow_check/demand.rs	5583	4d3fe12453e1c8ba970b776b467e001322dacbff9f3e16c39b9f5d14e060d3fe	0
crates/cairo-lang-lowering/src/borrow_check/mod.rs	13215	8b4e1d40b93b9677bb8f4b8fce1b2ae859f7649b77511ee11dcc39f3e78aeb21	0
crates/cairo-lang-lowering/src/cache/mod.rs	47940	b19a1e05de005add7bf7c9def38c140c697c9d5f0e668d891b30de502a1dd159	0
crates/cairo-lang-lowering/src/concretize/mod.rs	3441	0c6eecbf3076c362dea7c2f9cd0d14791803bdb71c1ce594123ade83c351bab3	0
crates/cairo-lang-lowering/src/db.rs	34220	af066a3e5c236536aaddf33bd128bca08aa3a3b8ff15e36238b127df5e4acd62	0
crates/cairo-lang-lowering/src/destructs.rs	23205	d6e853d7aec164de87632aa6ce285e009d425336565af3e688521b2f5edc55c7	0
crates/cairo-lang-lowering/src/diagnostic.rs	11376	0b435713634379f819c4333bf2e2a923d13cec3a4b9264af689d5bff17b5b891	0
crates/cairo-lang-lowering/src/fmt.rs	12196	59a177f69e262b87cd93ec3704a3b575a357be619a77b7a7a5fd595e09609605	0
crates/cairo-lang-lowering/src/graph_algorithms/concrete_function_node.rs	1477	6a55cf331291d860b32c110549a8bdda5624a7bd65bb6ca67d74e0efe17f1620	0
crates/cairo-lang-lowering/src/graph_algorithms/cycles.rs	3455	7749745db5f2c4713674c4a0235979531a23904da3903436d35fde4015fea736	0
crates/cairo-lang-lowering/src/graph_algorithms/feedback_set.rs	1949	dbc0b6c5f448f4a518d66625da90b1cd64be3dd18df91cc35df0c8e5c0ebf52c	0
crates/cairo-lang-lowering/src/graph_algorithms/mod.rs	109	80071fcfac46f961acb7f6803895b2551247f79ca12d467346bea6e81b77e5c6	0
crates/cairo-lang-lowering/src/graph_algorithms/strongly_connected_components.rs	4711	6fabe78161fed54ff70673c392a4676cbdd6d32d40b91a64bd78163b239f45d1	0
crates/cairo-lang-lowering/src/ids.rs	29037	540602eaede9f37a6641cf292e25f8f70c93852d74f4e408fffe75a5d46bac5d	0
crates/cairo-lang-lowering/src/implicits/mod.rs	13596	14392c0bdb472a52e3a2377351489e0130b393a89c7251097a6bfd02706c9570	0
crates/cairo-lang-lowering/src/inline/mod.rs	14922	9216e210ede3060deb0dab0e58dfaa7cd4b5ddf9f038d38e47200366b1b89bb4	0

crates/cairo-lang-lowering/src/inline/statements_weights.rs	4140	9447c76cdeed14932ecd804da36dbf69a81c7ddeffe5ac935b7b4cc7a20fe9a2	0
crates/cairo-lang-lowering/src/lib.rs	594	b84d1165ba24e0c4ecfca2c39029a7907b253bafa3c3aed31a2edb850649af69	0
crates/cairo-lang-lowering/src/lower/block_builder.rs	24323	78e2e096e120a396c5627ec16349feedfcb40ee3bdb5194be6000de0cd64e729	0
crates/cairo-lang-lowering/src/lower/context.rs	20019	0132d0571e95ac27ebd8da8d276e79bdde2c8a1da1f10a1041d45756980c2e31	0
crates/cairo-lang-lowering/src/lower/external.rs	1904	113ba121828e68edc3b44818ce641a13b744c7a8ae238c8d8a1cb8e3b95c0bb6	0
crates/cairo-lang-lowering/src/lower/flow_control/create_graph/cache.rs	1341	f0de5627e4a05cac42c852a76fd370a7e9e81423940425554acd88cd6118a931	0
crates/cairo-lang-lowering/src/lower/flow_control/create_graph/filtered_patterns.rs	5881	dd755be7b3df06c042c0a8ab3211d29cbdcb7e950e58fad011927b360c697cb4	0
crates/cairo-lang-lowering/src/lower/flow_control/create_graph/patterns.rs	42185	7a91e3f2276a94e2ba28991076c6fae7ebd1f93bc78f0947f82c6fb028271288	0
crates/cairo-lang-lowering/src/lower/flow_control/create_graph.rs	10966	be3372a9795ced5aca33701e141978c3c4fba413500f6d4e2bea2ea4fcdb2b83	0
crates/cairo-lang-lowering/src/lower/flow_control/graph.rs	18163	358bc820ba4be18eb2517a6950813993e34f1976a7132afd34791a3dc9ad5ea6	0
crates/cairo-lang-lowering/src/lower/flow_control/lower_graph/lower_node.rs	25960	1ccf7d7eebb63b5a06231cd3369cce4f6330c111cfbec94e3a16273f680537a4	0
crates/cairo-lang-lowering/src/lower/flow_control/lower_graph.rs	11142	b8e1a4a2d10142014311eed4de567ca2fe706acb983cc7966ccb61bced1a59f0	0
crates/cairo-lang-lowering/src/lower/flow_control/mod.rs	84	f476c1199d0a460d721205320a51d0baf042da8fa17ce40094f1766f3e7290c4	0
crates/cairo-lang-lowering/src/lower/generators.rs	10428	10ed00c9417e32c382e41425f5e3f0aecef7027670899c6e5ff59cc097c18703	0
crates/cairo-lang-lowering/src/lower/logical_op.rs	3947	1e6c50dccd144076624233641b69ccb8a53af2f4c48e2da0be6ed0660d25d48b	0
crates/cairo-lang-lowering/src/lower/lower_let_else.rs	3781	a077e0507a2259c128af6eb31247d0723fba24a589713708ee92a16065701d3f	0
crates/cairo-lang-lowering/src/lower/mod.rs	93437	3a8d54ef9d1287bcc9dbe78595b290ebad631bf56a8e9d42db10a504011e8af7	0
crates/cairo-lang-lowering/src/lower/refs.rs	19875	e203fec60a9f53178234d222f64c257aeabd19ab7851df8fcbc433953e578fe1	0
crates/cairo-lang-lowering/src/objects/blocks.rs	4492	a54bd1e77c6720a7e753e7a6fe68c8ba15c2383d5a404da351ab3e5cc9c86fff	0

crates/cairo-lang-lowering/src/objects.rs	23944	e71434d45e4d5400ac352cef7b493ca4bd7e88894421fdfe68236e1c9bf83cb3	0
crates/cairo-lang-lowering/src/optimizations/branch_inversion.rs	2446	da9b78a2ada7d8a22b907011a650feb3195cc26d174360e0016caa5a5f7b88a8	0
crates/cairo-lang-lowering/src/optimizations/config.rs	4299	4cbb181749bffa80e954c8828d6ceb60a9973aee8cf58ae3daa9f8c33b88b2e6	0
crates/cairo-lang-lowering/src/optimizations/const_folding.rs	89030	f963183d32de175aff907ee71c5a7f1a55d99a8bb775faa15ca76002b5bee8ea	0
crates/cairo-lang-lowering/src/optimizations/cse.rs	8554	8f5f9ebde40b99c431f36090b11ab7f6425ad399ea2f4eaf92ca38f7c37937b0	0
crates/cairo-lang-lowering/src/optimizations/dedup_blocks.rs	13329	420052777a004cfa0c5a8187bc2d9b4c37a9d05be9409a6de7ec1d3102ee1c11	0
crates/cairo-lang-lowering/src/optimizations/early_unsafe_panic.rs	5924	52d9b2632f15cb6b4e2d9052e059fad3af68190aae7cc4d063757614fb899cec	0
crates/cairo-lang-lowering/src/optimizations/gas_redeposit.rs	6421	c196e3c0bd06d08810a557952bf4ab67583a28151ecb7a37772f1c72f1b98909	0
crates/cairo-lang-lowering/src/optimizations/match_optimizer.rs	26147	3bc67741ca9fefc6bbd2ee1cc88549682e1490ccf5356cc7cc2a8cfc682145c6	0
crates/cairo-lang-lowering/src/optimizations/mod.rs	825	ceeb9e8c4b3096e5f9576ce3a36f0ec74028eb7d65b39c438581d5937291eea2	0
crates/cairo-lang-lowering/src/optimizations/reboxing.rs	12366	8e2b466f50686363878e66dd7391bc46a23e44c1c4a28109ba3a03d9af96ee66	0
crates/cairo-lang-lowering/src/optimizations/remappings.rs	6065	6ee8c0899529a617e65bbd2a7bd7f8ceff72a970dfaf9a501b4775b872bed374	0
crates/cairo-lang-lowering/src/optimizations/reorder_statements.rs	7226	de86fe440ba7baa6c105d810230d8f7768dcf01637deb7b05c440ec84f78d87e	0
crates/cairo-lang-lowering/src/optimizations/return_optimization.rs	22739	850c997146705977a620ebbcd7374948b423a6433a4db7f3de1afcfa8483662d	0
crates/cairo-lang-lowering/src/optimizations/scrub_units.rs	1692	a10124a0defa610021319c981fe4bad9520279ca9a8964c022fb42b1b6221171	0
crates/cairo-lang-lowering/src/optimizations/split_structs.rs	14732	49a74015128e3cd7aac0772c699a0f4091a9e3937b79d7006ec6c7082379538e	0
crates/cairo-lang-lowering/src/optimizations/strategy.rs	11189	860728c9d8269e1b716474f595e2f85a889c2df2bd70597a50a529e3dd91e77e	0
crates/cairo-lang-lowering/src/optimizations/trim_unreachable.rs	2154	361f8c624b623dc8482d814fbdcea2cb9d2442d13e5d65c4bd7bf9ff246147f3	0
crates/cairo-lang-lowering/src/optimizations/trim_unused_params.rs	4944	647e3bd580b2671e235bf774d07c0ce07899cd6942ea283325b109485f5495a4	0

crates/cairo-lang-lowering/src/optimizations/validate.rs	6179	e4f34f8471a8235a77a2f89adc067d95cc39b4a7fa9b48e9cc3b9816dd35bab5	0
crates/cairo-lang-lowering/src/optimizations/var_renamer.rs	918	4c2a03c8851bd67107a675be487cba3e215a55c68f13ae1ba6966cf38b7142e7	0
crates/cairo-lang-lowering/src/optimizations/variable_forwarding.rs	17948	d250e0787b533301470676ad17bb8a26f4b1dfabd55790d2f1eda7de054366dc	0
crates/cairo-lang-lowering/src/panic/mod.rs	21240	0f3a6701ca47707828c44409dc985170c3a80c304fa09f7e7967819b047fa66b	0
crates/cairo-lang-lowering/src/reorganize_blocks.rs	7689	88adf4b47aedd29d8359a403df49f601b635f967d3c84c0df0e240c379815c7c	0
crates/cairo-lang-lowering/src/scc/mod.rs	1561	8f08018083244d298bc855405a993e9b1c46fc9dd7ebacd3575c69b712a8c35f	0
crates/cairo-lang-lowering/src/specialization.rs	15258	16f24c164cde153dfc910ab5adf9694e08bc1dafa627f19d525c3f81b2e1437e	0
crates/cairo-lang-lowering/src/utils.rs	9139	ed1ab606f286a95365f148c40f1d576c5270a63a772ba702581d53fcc1a64291	0
crates/cairo-lang-parser/Cargo.toml	1210	7b0e396f29d57b10e4e08f530473daa91c60a0ac62284040c5c54165b3a6ca22	0
crates/cairo-lang-parser/src/colored_printer.rs	5599	70f83bddc163ffe17a19a42a3e692920ec8f765b946120dcc30b80e9f248d70a	0
crates/cairo-lang-parser/src/db.rs	4956	91f1529a768976f5326e8605115c58b1f5cf9f08938ffd4d92c241cb8c747437	0
crates/cairo-lang-parser/src/diagnostic.rs	14620	36d7be6ff02bdc192c58ad06b0b79526d3ce939435d5a9b914926db8d81e3b70	0
crates/cairo-lang-parser/src/lexer.rs	18794	5d7a62104d7d151d39783c4be9f2794a9de2cb548a023d6e97d12c805ad0feff	0
crates/cairo-lang-parser/src/lib.rs	345	1e3e2962b7509b32b19ada12f5a2ce05fbef68a100dde9b63a0c605d149c7480	0
crates/cairo-lang-parser/src/macro_helpers.rs	5747	eadb6e26b0706911288793ad07c09d1e4878558065784bc55bc568042e7e4df2	0
crates/cairo-lang-parser/src/operators.rs	1605	ade5622a3bfeb02e7d8e9a5f3ebff7920c75e0a4d2c00f13c940f1a8e12f6ea1	0
crates/cairo-lang-parser/src/parser.rs	178412	f19287523a912436719202fa40a23f27d99457f02dbae36f88a28d6126dd2d94	0
crates/cairo-lang-parser/src/printer.rs	10308	9e61c1b2607b3ad3ae68d63cfccd918dd6dbd73777e16c8f2c1da26a350501d9	0
crates/cairo-lang-parser/src/recovery.rs	2409	af5246d49b6fdb6d0ae2bab50b059073be1af014badac8f24ccdb8d10bf28851	0
crates/cairo-lang-parser/src/utils.rs	7015	533efaff1a845c84988bae6c1ac1d465b44165d08ad7b9fa691d24f9d1ccc164	0
crates/cairo-lang-parser/src/validation.rs	5575	0f2efddc5d698410af4a2754844bb2b70c724d066562cf8094915dd2eb4d1681	0

crates/cairo-lang-plugins/Cargo.toml	1140	f6efde4f4a63f369332f42a755918c9cce2f63e3a4c9ce1f363ba3d66aa83db5	0
crates/cairo-lang-plugins/src/lib.rs	954	23705319b162a9884a2352a75cce43541db32fa65d253045dcee88af12ab920f	0
crates/cairo-lang-plugins/src/plugins/compile_error.rs	2397	59251a537ca9ad984ff0e61791e4628cbe896e4e69a3f6ac8316e1f8a3f1f4e0	0
crates/cairo-lang-plugins/src/plugins/config.rs	12869	b750cf21ac3542b109ba18ad18b00476e15d0482a259c165af46dca03441dcc1	0
crates/cairo-lang-plugins/src/plugins/derive/clone.rs	2449	8b1d71ba200bfebb1221fd6eb2b04ae48e8c8b568c452c3b7c4669ab16074471	0
crates/cairo-lang-plugins/src/plugins/derive/debug.rs	2718	3ab06055acbbc525d25a8de854eda46a4089f0b82d09eb1c7d5c55255ac52182	0
crates/cairo-lang-plugins/src/plugins/derive/default.rs	3932	38a3681310c0d4592f310d21c3d91f7542be3967646444ab9d044aaf18d9d788	0
crates/cairo-lang-plugins/src/plugins/derive/destruct.rs	1709	c3972013edba7e9e0ec9a9ffdde12d711d6dd948f3222dbec2f0046e06ea9f32	0
crates/cairo-lang-plugins/src/plugins/derive/hash.rs	4235	939a38280ad36fbf266478a25613b515baaf842a668ebe7445e9cd0cd8eb412a	0
crates/cairo-lang-plugins/src/plugins/derive/mod.rs	6031	4ed43f5138e2fd8b5c0e027709e155991f1e6eecb824b678a0af29d0fb2356c7	0
crates/cairo-lang-plugins/src/plugins/derive/panic_destruct.rs	1793	f99fc58ae6e6c2fe1e965675305dbaf513dbab0978a9387ce85d8acd8fa340ed	0
crates/cairo-lang-plugins/src/plugins/derive/partial_eq.rs	2551	c9f8566fdd6ea012176ebd63c73080df4d931741dff1bff9baea0c9755f2f01d	0
crates/cairo-lang-plugins/src/plugins/derive/serde.rs	4043	6c14d7ccad7269bee5397268f00bebaef7733eae7dd1bcc293794ff7d1a736af	0
crates/cairo-lang-plugins/src/plugins/external_attributes_validation.rs	4616	cb924f8e58aad5dc56c0e5acdd58da52b9c8c25965910fd904f815a9ddb0f4a1	0
crates/cairo-lang-plugins/src/plugins/generate_trait.rs	12129	cf831bcb4644e9eb022d2dcb3b87ba7be4decb47f5064a6f966e9c19cb6ada8d	0
crates/cairo-lang-plugins/src/plugins/mod.rs	286	e634eb1149fc731452891ce7d041d959d5fe7179d46f572cf6e9eab51415e60a	0
crates/cairo-lang-plugins/src/plugins/panicable.rs	8277	c8f7d0a1e79351377fa47cee4c0fa9d057a6dd3afe974eb8368bf515890e6856	0
crates/cairo-lang-plugins/src/plugins/utils.rs	7882	9b1fa829203d29a29be80c3826a7e9cb895a4e6ed987fc9507887d7bffabad7d	0
crates/cairo-lang-primitive-token/Cargo.toml	205	977928aaafb88ef97175a4dcb2bedd4c54d88bbf7ebd3673e476c9cf60425f5d	0
crates/cairo-lang-primitive-token/src/lib.rs	1180	2d89755333baf08480eaea6bc8ed1772af47b91f5ff8338a358d99425a1a47ce	0

crates/cairo-lang-proc-macros/Cargo.toml	675	89bb0e1f708c4addfc60dc0910bcb11a1430036f8c04867282d3a460364b7b54	0
crates/cairo-lang-proc-macros/src/debug.rs	7570	ea547fe0b755b62048fefd6cbf5d9776a4e9657cc31208025e6d6fdec81ff702	0
crates/cairo-lang-proc-macros/src/heap_size.rs	4997	488dff258ae7c7746522ac134cee41b80f0430eb089247f2e4252ceb5291392b	0
crates/cairo-lang-proc-macros/src/lib.rs	3864	7037ba8a301c548c5816e3b87e4ef51b19edb22457ca3a160060e3a51d572d6d	0
crates/cairo-lang-proc-macros/src/rewriter.rs	7262	ba4a8fd2144357d6f74c96cf11f46cb631c9dc2f3277679037d9a9e24b47e1c1	0
crates/cairo-lang-project/Cargo.toml	596	c2075895152be51e923eae844c011aacbda96759ca87c724ad8e1da1d8b6a584	0
crates/cairo-lang-project/src/lib.rs	2750	00bf838cc4f53a1c192edf71f401fdc5fc9335fd3cb71e02669c217e344323bd	0
crates/cairo-lang-runnable-utils/Cargo.toml	887	13160c5c05a0dc51141cea307b87f9392423c479774d9953a1034c7ffb62b0b8	0
crates/cairo-lang-runnable-utils/src/builder.rs	28440	c6788a664274e4fec80425a149fe3f345941f833d71005edefe8b1535d9e7079	0
crates/cairo-lang-runnable-utils/src/lib.rs	17	981eec34c62f5531f29016aaaa8de2d750f4bc73add348fd0ba0340c383b19c1	0
crates/cairo-lang-runner/Cargo.toml	1711	0ce7f20d1ccd919dbd42bbf3500decb8dc506cff9b0bf0ea9f5a42ecaf475d6c	0
crates/cairo-lang-runner/src/casm_run/circuit.rs	9578	a567d4070bd5f423bc606b806d92f44125e667d4f28e4c0c8f4fecb2ee310968	0
crates/cairo-lang-runner/src/casm_run/contract_address.rs	1138	87e8dddfd65133178d191a2a69216362bb43e07e82d2c24c3aeb451b0fbaaa0c	0
crates/cairo-lang-runner/src/casm_run/dict_manager.rs	5095	8b5de79ee35f38f9818bf9c4a8440613f1abc7f90e93abb48b5b4fbabc8d11f2	0
crates/cairo-lang-runner/src/casm_run/mod.rs	104550	95c92cb2bb92a68fe0a43ca323e96c1554fade18bffc6179bbc4d22075a3e6c4	0
crates/cairo-lang-runner/src/clap.rs	795	d3193d4e1c0872e6f5b28053ab2780222cbfb692e4cc5ca251c6e17f36eaef63	0
crates/cairo-lang-runner/src/lib.rs	22867	d4c16c1ecd0a4a4467f59e88fab0e2dd97fce6d9f132913990debdd01af16d22	0
crates/cairo-lang-runner/src/profiling.rs	33152	9a123aae121cfd69630e3f46df6cfa03117bc5939a23b56affcb9e60a2b4db9d	0
crates/cairo-lang-runner/src/short_string.rs	1865	e8b84b28ba08e5ced7f95959e5e6d5b2544bf0510d4726ca9df8f96358c72784	0
crates/cairo-lang-semantic/Cargo.toml	1840	36759e06904053a1b94385544de86766f0a24978e4168ea076e93c3348563e66	0
crates/cairo-lang-semantic/src/cache/mod.rs	75211	2657a935118c4ba86d00e269e35ad25ce77d30b453c60e8b8d2bba5af4ead191	0

crates/cairo-lang-semantic/src/corelib.rs	49941	c60235bee4c60235ce8a03e0b3b4fd70ddf7313babf89f17d6183c7c5ac1655e	0
crates/cairo-lang-semantic/src/db.rs	27798	7b2c59e8e9525f25f817f41b330be3e688f5109673cd19ca428df1c810d911b9	0
crates/cairo-lang-semantic/src/diagnostic.rs	98540	c7fb4d4bd18844371dec6bdc0f8dbb8f9a2d900fc5515a670aa3b4f5b3f07cc5	0
crates/cairo-lang-semantic/src/expr/compute.rs	216662	9eebf2350067bc86ba420d527c206657455cafde1a7a8455b4fe993b68979fee	0
crates/cairo-lang-semantic/src/expr/fmt.rs	1439	d4b19b1f258c95339a9decd384c286dc2ccb3307ce31c49a5073e4192920a3fa	0
crates/cairo-lang-semantic/src/expr/inference/canonic.rs	24736	fd01ec1cb77ab0fe9852687e9c185bdd5f02fda9a308ae01ca9521c736c9ea6f	0
crates/cairo-lang-semantic/src/expr/inference/conform.rs	38910	a18f97d20506855ead0224e909edb019022d2e8a8605cec38e356b13395540db	0
crates/cairo-lang-semantic/src/expr/inference/infers.rs	23418	d183dfe2cefc48ef7e75dd2aa3301f8a5dbea740eaccb98ae0e38b9f92653b26	0
crates/cairo-lang-semantic/src/expr/inference/solver.rs	40404	1ef34a75a622f8a8cbddc6382b136f88637ed76a1e31ef9212cb903657a19426	0
crates/cairo-lang-semantic/src/expr/inference.rs	79595	70307c92bd3a0d208eccecbac8a05a045c346f45ab69eb7267ba36c436293f3e	0
crates/cairo-lang-semantic/src/expr/mod.rs	199	ac6f699225cd3b8ad1fc1b95655e8f49e2039991b198a7572b7df928eff117a2	0
crates/cairo-lang-semantic/src/expr/objects.rs	21703	c62fcd0440687f6d25c0c6812f3eb1974ea5120cc835f39d23e73fe575d77b42	0
crates/cairo-lang-semantic/src/expr/pattern.rs	10217	b7ce9ec9bf8129a6a2aeceda0b20c64229e32bbcc3b9f96bb6a5c6ab31f6d965	0
crates/cairo-lang-semantic/src/helper.rs	3549	b24a13f800a0e549f12b7fdb7c02e2971610b7a9dd254e7fe538fd494e5f8de6	0
crates/cairo-lang-semantic/src/ids.rs	1446	18ae7f13c6985c573c777ed6be89a827edf84c31737b0510f06a64b5302f64ec	0
crates/cairo-lang-semantic/src/inline_macros/array.rs	3385	3ee861083eb4c0c0a711218599e4465baa25bc5ac16ed865ffbde20d549e7ee4	0
crates/cairo-lang-semantic/src/inline_macros/assert.rs	6560	38d8ed1f9e76137760ce93116d2224fd6192ffaab41a859c99b77b0e5ffdb08f	0
crates/cairo-lang-semantic/src/inline_macros/consteval_int.rs	8942	5d2b042f235b3cd27b17acde6d6034d9ff8ed120abea8ab2b83ec626eb09c0db	0
crates/cairo-lang-semantic/src/inline_macros/expose.rs	1749	0481a3e653e1760080beff1fd455e4231c81d665f207810b584cd56241c87bfc	0
crates/cairo-lang-semantic/src/inline_macros/format.rs	3325	eac7e9ce1a20944bf038947a84368d133a4bf2ed0e15b0e6bca2f50bb6e99dc2	0

crates/cairo-lang-semantic/src/inline_macros/mod.rs	1225	ce634d7a6160137504c99727d72ea0e8358411b5148102971a9f7c6b61dec3de	0
crates/cairo-lang-semantic/src/inline_macros/panic.rs	6342	850b79c362476df34d568967a037eb7dcaf86a9c2e839f84dac82d989333c260	0
crates/cairo-lang-semantic/src/inline_macros/print.rs	4424	68a12b5e6b26970e75e4281291cf16d6b72fc8147e1e27e552be0a0f5569564d	0
crates/cairo-lang-semantic/src/inline_macros/write.rs	25627	13e16309123611cb911de31d4f312b045b0f5dc738d4d99696af4d1f53bafa42	0
crates/cairo-lang-semantic/src/items/attribute.rs	6487	5ef8754feb279b1aebee83e4bc2f310fdf78dd7b7b82d38ce0f25dbd2d19ab76	0
crates/cairo-lang-semantic/src/items/constant.rs	67872	50c20d530433dc8ad088a545e41b9ced5770b04d1e34c05b9409f2337594d0ea	0
crates/cairo-lang-semantic/src/items/enm.rs	14966	1f1db7768ced681b5439a15a01570b5dac99e34578dbe8fb4f0600dac8015c9a	0
crates/cairo-lang-semantic/src/items/extern_function.rs	9077	297998289db4222cc3f7795cba60172c387bd9f1c25e8389b3c6aca41c3037ff	0
crates/cairo-lang-semantic/src/items/extern_type.rs	6287	598e31bc23ce12c4408700a029352acdf87d5f11e09da7617836feba7e10afb6	0
crates/cairo-lang-semantic/src/items/feature_kind.rs	12988	f6433830f2d60c7ab0732fc1f1036c53b6e5412753de7a68b3594ad1fbd8034e	0
crates/cairo-lang-semantic/src/items/fmt.rs	2578	ef07997681439a8a6fb734663a79e640d4210694a1524e958f2c7d34dc3ca66e	0
crates/cairo-lang-semantic/src/items/free_function.rs	11734	5f6ec03b4630fa207159ed7780b8bffa35e6527e37bfcbd0b33d2f1168fae461	0
crates/cairo-lang-semantic/src/items/function_with_body.rs	13887	6b76f944ea39f918db6affb9d0e173888fe717570892665f0ea406a4ba540f86	0
crates/cairo-lang-semantic/src/items/functions.rs	47159	0c82ffe27a78b4148eda3ce0e99ec74c443f216e3d3175644972cf12605ab01d	0
crates/cairo-lang-semantic/src/items/generics.rs	36592	7ce2164f0484d60f52beba435025b77078f4c1eb0c687ffcfdaf4e2d111ff56c	0
crates/cairo-lang-semantic/src/items/imp.rs	181339	013653447500757fa415d093c16879ea5b11df2086074d241567798be75e2998	0
crates/cairo-lang-semantic/src/items/impl_alias.rs	11604	1be3d28274260d86dea3904d1b3a3f852867bafb99f0171041d0a100af3925fb	0
crates/cairo-lang-semantic/src/items/implization.rs	2114	5a4e386921bd8ee76efd68b2aaa63b162c1fe991b9ca144d7c54e0ca6ca64773	0
crates/cairo-lang-semantic/src/items/macro_call.rs	16093	ce81f586fffad4620d7b9a0be7569cbcd0fe679e3c38b12f99d1f90f64e76bbd	0
crates/cairo-lang-semantic/src/items/macro_declaration.rs	39356	b851ccb16c7726191b35f25df9edf65666f4b19f0d35024f60621121641a25f0	0

crates/cairo-lang-semantic/src/items/mod.rs	3343	bd532ee08e7b83187278354c395f983fb7b5d5bff87fa40b36e44c53bc30f8f0	0
crates/cairo-lang-semantic/src/items/modifiers.rs	2569	c7f08264e3a65f93d7dca167369056c37364bc4ead99e9e0cd8c95174363eeb5	0
crates/cairo-lang-semantic/src/items/module.rs	16730	1ccb97699a0cb53b759f5111f8945b8412f46454ffb7d84a2782e45df232f3fd	0
crates/cairo-lang-semantic/src/items/module_type_alias.rs	6216	b193a6fb37e194d2bcdfcc237242cd942122978cf4ab923408b9345f5b9df1fd	0
crates/cairo-lang-semantic/src/items/structure.rs	13071	127230b6b2f54a9fb97e8e241d1261662f95fd2c89d2cb73c25b74a88885acc2	0
crates/cairo-lang-semantic/src/items/trt.rs	62808	0191eabf5313d704c0d99105625b5f3a5f6025d4a0ef180ab781d042dab581bd	0
crates/cairo-lang-semantic/src/items/type_aliases.rs	4372	f846fdc5385e74ac7b8370480f98a73248f9d1a0bef183c9c37c585e3301d1fd	0
crates/cairo-lang-semantic/src/items/us.rs	20936	67c6049db196a45ee9dd3f912a6e9d7570ed46e223af75e9c0afbd87024e197c	0
crates/cairo-lang-semantic/src/items/visibility.rs	2234	dbb9f224a29774307955ca4574ff02fd813ab7f8d336310f699b34c9784e9568	0
crates/cairo-lang-semantic/src/keyword.rs	464	4fe452dace02195a18cdc7d80440d91f1d95da065c8a43132b2e5aae011309ac	0
crates/cairo-lang-semantic/src/lib.rs	723	5ec4561c150bc8c1bd5683980bdd51bf99f6503c5e34ce75fa8a9f4fa9d975cd	0
crates/cairo-lang-semantic/src/lookup_item.rs	10066	19d17989d9c45ef5f7ab15317069f891f2b5472cce404527d1ae896d1d463d85	0
crates/cairo-lang-semantic/src/lsp_helpers.rs	26444	0a87f230614cfb047ff6dad8ac29b530fe6b7b680ee0cf9cd118f1a62181440e	0
crates/cairo-lang-semantic/src/path.rs	25472	26103632724448a8382de03f4ed8a1edbe1d22d6c9f70bed20f18205442ae2a9	0
crates/cairo-lang-semantic/src/plugin.rs	3650	a7296359f53d1057b5cf9043b22a9a7bae23ab64b4a1e68c165ce5f87152e37a	0
crates/cairo-lang-semantic/src/resolve/item.rs	7160	fed332ec2933a5120261a0e99eda76294a3f0b30db1ecf3552bfbb490eb3f115	0
crates/cairo-lang-semantic/src/resolve/mod.rs	115112	f90f6bd352b33bf9a98105d7e27b57197524dedc2d8c3aa3ea524c4e8a7e23c0	0
crates/cairo-lang-semantic/src/semantic.rs	4881	6bdf91fb77f41ae90b94d07c771915e16b80ebe428967d198bb673df90ad27cb	0
crates/cairo-lang-semantic/src/stable_order.rs	13497	74f9cc767182f9fb9dec27683be128cc990471b17a7bed214e3bb1f01439b418	0
crates/cairo-lang-semantic/src/substitution.rs	31195	9b91cc49df741a83a84b417ada612e98165e42a610a532ab3b4ca341b747b3a4	0

crates/cairo-lang-semantic/src/types.rs	60452	5dd49c5dc049efb3118d3939833e7b4d1b9c7e9d44fb6ed7a87c459aa1536536	0
crates/cairo-lang-semantic/src/usage/mod.rs	19323	78b5d62e717d7ae881238318284278c99c7aa085e6856c21e5bce08d912ef727	0
crates/cairo-lang-sierra/Cargo.toml	1245	6da6c0ad10e09b2fa3f650b9e064b6ee05f2ee169c78db55bc86ffae24f42cfe	0
crates/cairo-lang-sierra/src/algorithm/mod.rs	27	635616f226b407b7a76f148a524510171a12788a88881c22905148868311e535	0
crates/cairo-lang-sierra/src/algorithm/topological_order.rs	3750	9b68085c090bff1854fc79857629377c6a12c927ec14772506dc4354da9a93a6	0
crates/cairo-lang-sierra/src/build.rs	71	dfc95faf6971ef88db3bde5eec530a3c7300c0486db2eabc195f3bc4f43dc80e	0
crates/cairo-lang-sierra/src/debug_info.rs	8426	3b6d52aae041247085268ee052fe7a6ec5a911cda952a39b07f389aabb129814	0
crates/cairo-lang-sierra/src/edit_state.rs	2100	ef9b611aa1494bc5477dfd7e530285b41f14434ec4b1f65791aff84672959f18	0
crates/cairo-lang-sierra/src/extensions/core.rs	6180	12f0f150c7502f6a9e0d8a568c1ef119a27c3c730ec10f2bd54b82f741df7c9d	0
crates/cairo-lang-sierra/src/extensions/error.rs	1712	0c2dbdcbd896148ad49f322ea9b67da8559fad4fccdc1f9ce09931ef0f544fdb	0
crates/cairo-lang-sierra/src/extensions/lib_func.rs	25513	a0ce27d0d8152f7866332d2637d896dd74ce595206d2248c16bfdec142a7b35f	0
crates/cairo-lang-sierra/src/extensions/mod.rs	2768	e979a5cf79be22f1d3b8c51700e56cfca8bdaec63a46a03162cdbedb8e9ec924	0
crates/cairo-lang-sierra/src/extensions/modules/ap_tracking.rs	2382	3e32039a5d495c6834359dbd6547ea9cd93d7bc6fe841163851d6746fa0d6988	0
crates/cairo-lang-sierra/src/extensions/modules/array.rs	26205	58857c3f5c6e0287dea3dfbd80ab9a882449b65e07357d86a3c11dba27fd47e1	0
crates/cairo-lang-sierra/src/extensions/modules/bitwise.rs	429	2c065f36227f4da6bea2f9dfc4f3217a2a9ef54ea32b36687169968495939606	0
crates/cairo-lang-sierra/src/extensions/modules/blake.rs	4770	aa655fa86edbfb74f57c244dbf8baa978321eab2cb371fc4f074dd21ef6cfc64	0
crates/cairo-lang-sierra/src/extensions/modules/boolean.rs	3780	fcde0ca79f812dd763504a64825dd789c1e2a6b8c65ad699a143608d50d9dc42	0
crates/cairo-lang-sierra/src/extensions/modules/bounded_int.rs	27956	d975e9338bb20b577f68668e889834d8f0978b4187a8a096df70cb948c8528a1	0
crates/cairo-lang-sierra/src/extensions/modules/boxing.rs	5284	84f669b7bfa932e686586ca1b2c1b604c8de1834e150b48d100caf3b42f7c7ce	0
crates/cairo-lang-sierra/src/extensions/modules/branch_align.rs	743	f9093feda987fa961ce60a7f849766d234ab9f93f4ca3e7e4c48e306c7c4963e	0

crates/cairo-lang-sierra/src/extensions/modules/bytes31.rs	2490	5befd20c5c674587804b97520480c6339f0ab588b1cb44a96bdc4113174de58e	0
crates/cairo-lang-sierra/src/extensions/modules/casts.rs	7024	c74e099a84afdc29e5be387bb926054e43ad7f3de379dcd95d8c450112553e11	0
crates/cairo-lang-sierra/src/extensions/modules/circuit.rs	49281	2d7ec86291a5fade010a898144d272967c982423ffeb9f779953b6fe88d948c0	0
crates/cairo-lang-sierra/src/extensions/modules/const_type.rs	16095	620be418f673af06518e6d109e08ded63c27df92c43f6a9d6aaa00e4bf19b7f5	0
crates/cairo-lang-sierra/src/extensions/modules/consts.rs	2554	1f02ab96b7b81f22697b054c88cfc14546a8e1adf9e66f5a1af9237f40dbb855	0
crates/cairo-lang-sierra/src/extensions/modules/coupon.rs	5850	cd80877e7d0a39c7f42070a90f7497be533e3c277ef786424b0adee7a094823a	0
crates/cairo-lang-sierra/src/extensions/modules/debug.rs	1212	3867fe6c80209032d80cf22a74a99b879d6b7b04abe34899427e67d3efb66678	0
crates/cairo-lang-sierra/src/extensions/modules/drop.rs	1232	a3405fb27939b034641dd54649e2fa3f96fff749bc014efdff16c2bebe0143d3	0
crates/cairo-lang-sierra/src/extensions/modules/duplicate.rs	1278	7c838ab049544ce49b84574d8edf643909d34841073f203158724528b016ed55	0
crates/cairo-lang-sierra/src/extensions/modules/ec.rs	15537	e4a18cd51071a02258d67b928ecdd63e765bd67d6fd9c0effd96901871585108	0
crates/cairo-lang-sierra/src/extensions/modules/enm.rs	17169	5df91eec8e6c98e7e1631aff88b8738ba0cfda41c80cc4e7f1f08f0a07196021	0
crates/cairo-lang-sierra/src/extensions/modules/felt252.rs	10120	d662fd68648b4bd5a86b2e477bbb57111f0783221b040db1c95be9b71fb9fed2	0
crates/cairo-lang-sierra/src/extensions/modules/felt252_dict.rs	11355	283c07442e2ed388d80bb3780891547bf96d9a02085b6878d91b5ce3e23e4baf	0
crates/cairo-lang-sierra/src/extensions/modules/function_call.rs	6553	0e6d0eb128810df637cead6afcfe5d9f19247765aaddc3bd41f95c12c2675db8	0
crates/cairo-lang-sierra/src/extensions/modules/gas.rs	14216	172514245ccc478545ffb5a732fd7da0eb2243ff85438e6eac43d02522d68168	0
crates/cairo-lang-sierra/src/extensions/modules/gas_reserve.rs	4478	68db4423be4185a3c5cb4bec4df7bb1f535456b56960c17d0002e9fde767d0d9	0
crates/cairo-lang-sierra/src/extensions/modules/int/mod.rs	7599	9d7af18faf9766f7606b32d5cc33180f0775de041b0a126b1a1994a7b1b70e36	0
crates/cairo-lang-sierra/src/extensions/modules/int/signed.rs	11796	c082c8e36d63c71ed1bdb57d8cb027cfbd262e91f5d9d8ed26691939ad5c423a	0
crates/cairo-lang-sierra/src/extensions/modules/int/signed128.rs	1530	3ce148e0c3b1a40012678f28db2ba9bf3167eeff7bfd37514dd7c2ec6cf7acd5	0

crates/cairo-lang-sierra/src/extensions/modules/int/unsigned.rs	15201	2a75a977fc80d9a78df0edd65aee4766c309652fbc57abd3b833eb02e9dea4a3	0
crates/cairo-lang-sierra/src/extensions/modules/int/unsigned128.rs	8767	4775854d8f3ad67615b08ee902b3703efdaf0543fc37fa98d6c10cd95ab834b6	0
crates/cairo-lang-sierra/src/extensions/modules/int/unsigned256.rs	6796	ebafd86d50b98750940da6ce1bb6646bf5846ea5d17828a6a6b1b6588ba18296	0
crates/cairo-lang-sierra/src/extensions/modules/int/unsigned512.rs	2951	b9690b437bbd86f4b1b1899215f3821c4b9f4be5c32ec507b3b1db934b3d2748	0
crates/cairo-lang-sierra/src/extensions/modules/is_zero.rs	2055	e0ba001d8b5d00882db8531b12fcda3ae6809889189a0b91a4107281c939cd33	0
crates/cairo-lang-sierra/src/extensions/modules/mem.rs	5730	57b1216779b3956cccb893ad49c8eb5d802ab8e52bea6db6bcab59e28b452031	0
crates/cairo-lang-sierra/src/extensions/modules/mod.rs	2307	5c343d1bed023e3bfbedd528f516865d46c05e9b2213d601bf7d4c3f2e54c712	0
crates/cairo-lang-sierra/src/extensions/modules/non_zero.rs	2491	51ad34a10e5558c7a2dc943f73474090f579359899133b995e632cdecb26bbbe	0
crates/cairo-lang-sierra/src/extensions/modules/nullable.rs	6219	69ef87ec759089f92563da902949a59fe5c38f157935d9eac24b020f8c42a1cc	0
crates/cairo-lang-sierra/src/extensions/modules/pedersen.rs	2142	76d48e9980b9df519671b5816f345a6718c770d94bc59b932cb1a17888eddeef	0
crates/cairo-lang-sierra/src/extensions/modules/poseidon.rs	2402	64ce45d4428488b0d68ee1ccf3f66c0ae9baa41b9ee5bd100c103c29d0d354e3	0
crates/cairo-lang-sierra/src/extensions/modules/qm31.rs	10657	37b1898865b7c471c029a72d267887c2d42a0198fa658fc8f7d4bfa7022d5f72	0
crates/cairo-lang-sierra/src/extensions/modules/range.rs	6940	d77235507ced078517487df038448881356159582ba34284c4b83a42d54cd9e4	0
crates/cairo-lang-sierra/src/extensions/modules/range_check.rs	784	af4f9d0735fb8ffb1b68a280953d810bedde54ab9da44b4d3c0c60e38d5e7e5a	0
crates/cairo-lang-sierra/src/extensions/modules/segment_arena.rs	1094	709ed651bbafec9c427297d1d01ac3a6511372a99a3ab116feb7185b226dda21	0
crates/cairo-lang-sierra/src/extensions/modules/snapshot.rs	3174	06e8993661dbc615c40f1a2b3f43be703cd36a7f744808aa1cdfdce4f610545f	0
crates/cairo-lang-sierra/src/extensions/modules/span.rs	1151	d2f49a2faec4e3f8b327038338aa7fd1ddf842641ef11f150b003ccaa1ee9a41	0
crates/cairo-lang-sierra/src/extensions/modules/squashed_felt252_dict.rs	3361	c3504bfe2316a7042ff55882b594830803295cf3fdd65acda4817cb102641e55	0

crates/cairo-lang-sierra/src/extensions/modules/starknet/emit_event.rs	880	549e50cc6af3ef5cdd4f5666d92d900907a647bd911c645270c69c443644fe8e	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/getter.rs	12948	08be307df3478990035bd150c89a492b4df5de3da5593343114a9d3c50b92482	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/interoperability.rs	8705	b6aaf1652db2e6e03ea43565e3db3b6900efef6929634b8acace63f2511e23a7	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/mod.rs	5652	74c57ba011fd1453e1b686f5d9dd36dcf7427f01dcc66443f11191d87739a611	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256.rs	6735	f82f3b08cb95fbe11b3b44b103e93b0fc66b2b7a64ea416213e1ca07ac07bfc3	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256k1.rs	1001	f70b17f0e1a3712c435d57454dafcc965c5ec630d25dfc4eb20eee65c69ba357	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/secp256r1.rs	1001	ea55c773bb58e7acfcbceb07facbcc6bfa84c30ce09fe1f46e8b37ed5caf13f2	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/storage.rs	8377	9b2f35658e7435a320bf653102690b7f8743d1f493b0e5bb2a47043a7e76bc6a	0
crates/cairo-lang-sierra/src/extensions/modules/starknet/syscalls.rs	14296	7afa8b0f441dd713716c42ee4ccccbcbed9bcb64e39c4f6812243eb108e95a56	0
crates/cairo-lang-sierra/src/extensions/modules/structure.rs	14362	1431e0fcec16fcfa80d47dda8d054867fd4556b99c41c04c7d7d009351175a7e	0
crates/cairo-lang-sierra/src/extensions/modules/trace.rs	1282	b799e19357f5b970d824f3c40173aec91ab60c2fbdaf200137b3b16de6c3109c	0
crates/cairo-lang-sierra/src/extensions/modules/try_from_felt252.rs	2479	ba43ddb6eb5a4716c1e5f27ee82838bb288e00fc5caed2c924658afca204fab8	0
crates/cairo-lang-sierra/src/extensions/modules/unconditional_jump.rs	813	6d6f93b0dd5be172a5ac2701344f7a6cde02d5d47c48964ac2c3c9ff03544a3a	0
crates/cairo-lang-sierra/src/extensions/modules/uninitialized.rs	1154	e6d3e7aa67958e2389be9f1de38b22a84638023f029c00d61c617d05dc8aec7f	0
crates/cairo-lang-sierra/src/extensions/modules/unsafe_panic.rs	706	27213b591fa1666788899d9c9d80bb06269b8b9f4817277f30aad07d0d65cab3	0
crates/cairo-lang-sierra/src/extensions/modules/utils.rs	6328	a7a62abfeb62c8c0d7f3078a821071a6653b1e578353417e8c726eaa15286e9d	0

crates/cairo-lang-sierra/src/extensions/type_specialization_context.rs	616	6eb6407ae435a7a68259420c00a806d4c062912345099b88ab90328d8f7ced21	0
crates/cairo-lang-sierra/src/extensions/types.rs	8638	d3c7e74495e5dc6326498aab23a6e1ae64f0ad9b62e3a2ec7fda9b5b3ab226e4	0
crates/cairo-lang-sierra/src/fmt.rs	7634	880f3d324f13c8720d5d624678915adc970fab53c016085dbfc8473b3e5ba824	0
crates/cairo-lang-sierra/src/ids.rs	4066	060c9c53bd269e24de6c90d317c3384b0b0db27ba4fd8c2fdc5146cd1d00f40a	0
crates/cairo-lang-sierra/src/labeled_statement.rs	3373	163976370406399e19804ef2253158fcc180e74625f658ec6210ceb8fd8c4d8d	0
crates/cairo-lang-sierra/src/lib.rs	1187	0e7f1d9b8c48089d879e2e333ad324e678dfc454253113b29e03eed0e4368ed4	0
crates/cairo-lang-sierra/src/program.rs	11121	a623d6eff63ed0994ac4fb5b4076d95880afc0e9156e5c60865e31398eb04bb6	0
crates/cairo-lang-sierra/src/program_registry.rs	17518	df1a6c14e19ff6d0c96e9985dfb5daa869afd60605c1388a3fcc290542c68a66	0
crates/cairo-lang-sierra/src/simulation/core.rs	33654	6674a9db10d3103dc87a89e46d25c4d0b80dbbb36e6a88b0bc2ea75336d45913	0
crates/cairo-lang-sierra/src/simulation/mod.rs	6015	1e742c8715edfe3b3111411209c8b02294e7c24b13100d18b64295909c611565	0
crates/cairo-lang-sierra/src/simulation/value.rs	734	3dc76e0ab0f29e0377dd547bbc1b604a42250fa316189baf768813f76cea1cfb	0
crates/cairo-lang-sierra-ap-change/Cargo.toml	764	c256bc01e51113db81d81a7c992770c596c5868e9e4c885dd74df2dbc8cd9268	0
crates/cairo-lang-sierra-ap-change/src/ap_change_info.rs	1108	6b60014bd489270fc85a15996accdd093b551614a1142118d2ae3143aeb19879	0
crates/cairo-lang-sierra-ap-change/src/compute.rs	14627	e821f794a050e9e2269ff4638e818f639a2a11a973f8211186310fcd594dee0e	0
crates/cairo-lang-sierra-ap-change/src/core_libfunc_ap_change.rs	28543	b177577c0c6dfcd9e5270c469952a60740138411084aaa0105f30191ffa23674	0
crates/cairo-lang-sierra-ap-change/src/generate_equations.rs	7565	bef7f8441314d81daffb80d695364b6da54ae2c3af3a0c364cbe7bce6e2dc8f1	0
crates/cairo-lang-sierra-ap-change/src/lib.rs	5075	47d6d60b61ae21a30a20b44da94269d5469a3eec1928bb5ac6dc62651e72fef0	0
crates/cairo-lang-sierra-gas/Cargo.toml	923	776cc01650253808ccc98e9ed1608f01a334d11b1428b557da709c54b5acce99	0
crates/cairo-lang-sierra-gas/src/compute_costs.rs	36226	10f38295ddf6515479a5e296f9fefcffce37fce1caad1e07487ccec421626c23	0
crates/cairo-lang-sierra-gas/src/core_libfunc_cost.rs	1993	f650dd5e2973da809018bfc7a9484eab8164326637aaadfb23429705bd7a5f3f	0
crates/cairo-lang-sierra-gas/src/core_libfunc_cost_base.rs	47737	38f0fb987f353da00e402943ee3d9700a86a59e06d20c7273ff1d904f2d6c0ab	0

crates/cairo-lang-sierra-gas/src/core_libfunc_cost_expr.rs	3068	f5224549eff968f358e87c21f030af438b0d6f5920b908d4737cb98d76b4d54d	0
crates/cairo-lang-sierra-gas/src/cost_expr.rs	971	70a854d5b5c545ae8bb22a9296f826605759bc57beabda29e8a1b2bbe2af1e49	0
crates/cairo-lang-sierra-gas/src/gas_info.rs	4793	3a8bf4e33b70532ba47de755759ae8618eba021b0710c85e4fc12acca36d07b2	0
crates/cairo-lang-sierra-gas/src/generate_equations.rs	4897	70f7904c4e67c38b816f489cf400a07e85cce85a4c399ba677cdeda334608b5b	0
crates/cairo-lang-sierra-gas/src/lib.rs	16437	4f8f46b37b13092382ffe15f93136da42ff43f0953c2f95c26c256bb6b701413	0
crates/cairo-lang-sierra-gas/src/objects.rs	5491	f9e472e895b22ab9c7222ddc962ade1133c1d73704847b216de30d0f79e3052d	0
crates/cairo-lang-sierra-gas/src/starknet_libfunc_cost_base.rs	4871	08ef9e2aa171bbeaebc229d784387a00f475155a690310bdeee9e0a869e9e053	0
crates/cairo-lang-sierra-generator/Cargo.toml	1907	2b139c3e1340cb3c0e78c8021cc432674a7e33498fea4a6ca37787bba2d8ac6d	0
crates/cairo-lang-sierra-generator/src/ap_change.rs	886	06770faa7c040d81cb929c80704e5173d4502a49b060e29a652477aa57550ad0	0
crates/cairo-lang-sierra-generator/src/ap_tracking.rs	6177	720c2e182891e8401115ea80872ad4c45caad28f93768fd15b86c4539283ec23	0
crates/cairo-lang-sierra-generator/src/block_generator.rs	31097	8ddbd2678b7a5330bb31fcfb360190e970a58cad4e5f1d8ed58748162d1b555e	0
crates/cairo-lang-sierra-generator/src/canonical_id_replacer.rs	2557	8a6c5d15b3f898ad9d23f9a887940ec915c45a884940f85b4ea05473f89b2ac4	0
crates/cairo-lang-sierra-generator/src/db.rs	20338	7855cb358f28f73beb12f6420266649c0b7a7c06655d43d15bef3789c455b8fc	0
crates/cairo-lang-sierra-generator/src/debug_info/function_debug_info/serializable.rs	2542	1822da65544d402d0a95b5a1c4767d7cc04593cc62d775ec31f0671579fbee53	0
crates/cairo-lang-sierra-generator/src/debug_info/function_debug_info.rs	23491	e5faa7ef511ff77c8398c0ec920457b80e6aaa619b7fbca4f224cc37ca01ded0	0
crates/cairo-lang-sierra-generator/src/debug_info/mod.rs	3116	a58306571f877f52c55d2b3359711eb716fa0b748e61eb8edd25c16d1bbdaa0e	0
crates/cairo-lang-sierra-generator/src/debug_info/statements_locations/statements_code_locations.rs	1327	9d7c37340f33502715e68fe85fd9ed29937cc771e3f5679777e34875ce838e9b	0
crates/cairo-lang-sierra-generator/src/debug_info/statements_locations/statements_functions.rs	7285	56bb45cf9e8e67cd4a8efb8bb848e3e95aa6b5e7a5efb40ccb59b79eda9b2544	0

crates/cairo-lang-sierra-generator/src/debug_info/statements_locations.rs	4898	b98770f292a5178a497c2c72f91913963a3185ef8a1dfe87d3e55037d4983914	0
crates/cairo-lang-sierra-generator/src/debug_info/type_names.rs	9028	1020d9f056137423deff4eade77c6993154be7a756e4acafa666f4864a115d7a	0
crates/cairo-lang-sierra-generator/src/executables.rs	4940	ddd4cbf099fcf21e66c1d55270e8f558420e265d473f09869d5392f71087fe34	0
crates/cairo-lang-sierra-generator/src/expr_generator_context.rs	9927	c564c130ae08b7d1c8660032c926dc6c5265ea002a7d5316fc1738d4e2718bab	0
crates/cairo-lang-sierra-generator/src/extra_sierra_info.rs	825	2ce4a506e83bc4589a43ce2cad0e792ec80c23b5a51299766e4f51818e8eff63	0
crates/cairo-lang-sierra-generator/src/function_generator.rs	8852	c273ebf8b59c9899860f8bdfcafc567c34585a3a8fc5989a4b6c7be7c6d629af	0
crates/cairo-lang-sierra-generator/src/function_generator_test_utils.rs	2081	f7ae0d5d27adcd4f88ce655b6d86b33bfc07659720398d511c57caa9c4370829	0
crates/cairo-lang-sierra-generator/src/id_allocator.rs	304	e0e8ef751ebc59bbf12ebbf8783b020b48c9d6cafba4435eb85a9211d876c009	0
crates/cairo-lang-sierra-generator/src/lib.rs	702	c5043b9ef18b32ec91085543a38e17e672ba1bd7ecb36f15ec2702e68ed59ed8	0
crates/cairo-lang-sierra-generator/src/lifetime.rs	10318	fd2afc2ec06d45113b72732e5cff2f7cfd4d5187904be23cfef0146e7a630cfc	0
crates/cairo-lang-sierra-generator/src/local_variables.rs	19715	edee41829621a172824da9795b5fb9679a0c2522625e13a41fced5cbe9161b9b	0
crates/cairo-lang-sierra-generator/src/next_statement_index_fetch.rs	1900	43bd8b05acb6781c9348963e6292f87f5b2cf0e5a97f02b1d381aebc2a3be2ca	0
crates/cairo-lang-sierra-generator/src/pre_sierra.rs	5826	1158fa380f4ae04d546ebdd3d95a2f95ef51993015ab6c0390f1edb336dfee1f	0
crates/cairo-lang-sierra-generator/src/program_generator.rs	18258	dba4aeaf736264f736d2b91f3264de5abe1c4b18eb553b632f349218a8d90a3b	0
crates/cairo-lang-sierra-generator/src/replace_ids.rs	8790	67df8f245c5a65cc484a1154d0ab6d8902864de6c007e936a65404ca2ab1af85	0
crates/cairo-lang-sierra-generator/src/resolve_labels.rs	4078	e744e0badaccc514d36bf9bdef6ed3466d9ec097ec888297c7b076e0d91b34cb	0
crates/cairo-lang-sierra-generator/src/specialization_context.rs	2235	9575217c3536ed0e1b8bfcfa93eaf4ceed8d264366d6a3461bc320f8882e484e	0
crates/cairo-lang-sierra-generator/src/store_variables/known_stack.rs	6986	ef9501a884d1ae60a22092d9eecd7abc27c9688d8d9c1487480ac084e9e3e378	0
crates/cairo-lang-sierra-generator/src/store_variables/mod.rs	25303	3b295e91b0571f98b89f0d2667c9c02ae7ef4eb4e58ef8cd3b7dc8a7a1946c53	0
crates/cairo-lang-sierra-generator/src/store_variables/state.rs	7846	e94c9d536cb09d5107d00c2e813c6e61e01a6ec384141aeee2e7c5a06463d203	0

crates/cairo-lang-sierra-generator/src/types.rs	11635	7f0d066312cbb9c4d674745adf2b2d571e66663fec4a773bffcfa2e1edb67915	0
crates/cairo-lang-sierra-generator/src/utils.rs	20935	d0bdd34307a0e9e353c2e13acf085ba0e5b51160da58af2f14432dea13a2927b	0
crates/cairo-lang-sierra-to-casm/Cargo.toml	1283	47b9bf575d4e324120770b608e0f2efe131a66dbddf161cd33a9708e0c69da1c	0
crates/cairo-lang-sierra-to-casm/src/annotations.rs	22289	06c81b0cafa2b83981c8e0a1f1e7e30bae6dd07fdb003f60484ae1730b2142fb	0
crates/cairo-lang-sierra-to-casm/src/circuit.rs	1229	fcffe45b5ef74d257a389035f1bf7ea4e57bd5eecd84d45dca09db53eef53544	0
crates/cairo-lang-sierra-to-casm/src/compiler.rs	30215	522cf0dbbd91b98908869825591bad9a6d018323f3216012098482174383a08d	0
crates/cairo-lang-sierra-to-casm/src/environment/ap_tracking.rs	616	2272a16ed69e1c5c46efe6d1f8513084d37be31a5979116fb2fd9805cc07a1d0	0
crates/cairo-lang-sierra-to-casm/src/environment/frame_state.rs	4596	0ab90267afa318932ebfa12d5f2ed21a21c458d37951d9b769677a7d127e79f6	0
crates/cairo-lang-sierra-to-casm/src/environment/gas_wallet.rs	2105	b6a75149eca42f911674f17b3a8d4777eb8fe3c3d02393203acaaff665685137	0
crates/cairo-lang-sierra-to-casm/src/environment/mod.rs	2584	0a911694273206a60eab82e02800137e8d5b4eb32727b5b67dbe9827fe753256	0
crates/cairo-lang-sierra-to-casm/src/invocations/array.rs	20140	4aea928916df27b75544bdf8e32024133ce0d19571afb7a0b351759c6a733acc	0
crates/cairo-lang-sierra-to-casm/src/invocations/bitwise.rs	1229	d51477ade4fce34c37bf3c9f5a4c245b15627a29bca94110fe59d969a6880e98	0
crates/cairo-lang-sierra-to-casm/src/invocations/blake.rs	1914	8aca5d84230addba16cd17de3a36390c8742003aaaef38d0c15665d0af670833	0
crates/cairo-lang-sierra-to-casm/src/invocations/boolean.rs	3251	775505bcc7867d91a2c3fc764f2a6cf05f6a1977c7fb43786d1cba93845c59d9	0
crates/cairo-lang-sierra-to-casm/src/invocations/boxing.rs	4097	8e54f54a8ebfdfde62f97418096eb4bf7c9a17773e0b40fb3296ac05d5868080	0
crates/cairo-lang-sierra-to-casm/src/invocations/bytes31.rs	940	9cdb770b0670b77c3364db2c824aece01ef9383455dc61e41f0d33525efa2955	0
crates/cairo-lang-sierra-to-casm/src/invocations/casts.rs	9079	c739825160e22ab33135211361cd9fc64944f89eccc31a56e4ced4fb3c201760	0
crates/cairo-lang-sierra-to-casm/src/invocations/circuit.rs	23950	436195e40b5b63e50a8e53ce03f6f20f55e7026173c8e998694482cd32f9e724	0
crates/cairo-lang-sierra-to-casm/src/invocations/const_type.rs	2512	ef1a24e051a0b694543b2a979ab2511a1cd459ca5ca819b4dd3782152bea1804	0
crates/cairo-lang-sierra-to-casm/src/invocations/debug.rs	1299	d35e3fe54d20b1bd69dc2644e9a7af5f716f3cbcc473a74779bea6a9e62702c9	0

crates/cairo-lang-sierra-to-casm/src/invocations/ec.rs	16612	a15f1d89b8045571643e2ff737b6c3fef5ac70f324c2d9cbffb482f852ddca72	0
crates/cairo-lang-sierra-to-casm/src/invocations/enm.rs	20273	9329d25f4d185b9117af3c35a5ce5c252cbaba3b5d47a3a8b8ddfd4865b79257	0
crates/cairo-lang-sierra-to-casm/src/invocations/felt252.rs	3992	ec63ad1f7cc35827639011f34cca26796cfc3d5c870274f75328558daed3b747	0
crates/cairo-lang-sierra-to-casm/src/invocations/felt252_dict.rs	34762	44d344bba719872a599fd73167947f9a5f299f515277286199266d22035faad8	0
crates/cairo-lang-sierra-to-casm/src/invocations/function_call.rs	1726	146f1bd5558bd7a16b73e2b64d36eedb829cab5c536fe51fb1a1455470c47a61	0
crates/cairo-lang-sierra-to-casm/src/invocations/gas.rs	12880	33683e857d4103801b3e52ae0844e2ccc805f42a1c6a326347045e8b0c8b208f	0
crates/cairo-lang-sierra-to-casm/src/invocations/gas_reserve.rs	1848	aa9244587acf95abe808d6b37cba1710338aa82dacfbac389c111cfbac065fbf	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/bounded.rs	13336	10ea21047b19bed493e13c02ecd1c258813ce028ed8b218fae04c1fa040076e1	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/mod.rs	5810	162f4afcc9dab40c0ee01f5c0dfa48be905bfdfa51e99279c06dfeb6c90d7c0b	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/signed.rs	6707	e37e3610c8f846b872a899e3491b6240ce135d28b0c864ad6ce1e42fd9498272	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/signed128.rs	1081	0e6d441c1f95af91ee11fca7eda31f01f8a2215fa11f5efa1c9d3f36d0ebe250	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/unsigned.rs	6742	e34bc55fb3b0806ed0dc70b54e108505ccfc1388e4f2ba8b3a1e1f6e2c80cb30	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/unsigned128.rs	16361	ebcbd7b7c0f00386ee8692c9b91377e79164390c611468762f5953d59f99ad3f	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/unsigned256.rs	22425	cadf78a5c6351cc79cc7a14129990ef78a21e55b93651358ccab989ebf47b500	0
crates/cairo-lang-sierra-to-casm/src/invocations/int/unsigned512.rs	8351	9c671e0073fa695cd8841002e6c45aff7eb594f6f50b1f2803cebef1cb562441	0
crates/cairo-lang-sierra-to-casm/src/invocations/mem.rs	6406	b9a48d9e7bf8113819d43465bde6e8db5d4f9c842422c97bc5ec5b8945dc31d8	0
crates/cairo-lang-sierra-to-casm/src/invocations/misc.rs	11998	38b4d607e5034490df23fcb069a3d49cc2ae065e00d6f6a9b4f5847667894b78	0
crates/cairo-lang-sierra-to-casm/src/invocations/mod.rs	33234	627df5e90a44cd2c8ae5f826b2139db1c2b304b36598135f41513847d53f275e	0
crates/cairo-lang-sierra-to-casm/src/invocations/nullable.rs	2132	fff675671e7e1db055d6f044ba099386baffed86d49f21a809365de5065e1d75	0
crates/cairo-lang-sierra-to-casm/src/invocations/pedersen.rs	1374	c80ee865cc8aca9e24bcdc73c0aaa34dd75987c0941e44870f113de91c2cdc52	0

crates/cairo-lang-sierra-to-casm/src/invocations/poseidon.rs	1476	c8d3efa310a6e3a90b827a01a78ecb061993dfe928a4d41d8df95206e3477561	0
crates/cairo-lang-sierra-to-casm/src/invocations/qm31.rs	6768	cc48e6d879ceb7ec85cd69d19ac735bcc7d3cc6ed76b143fa567b4ff0e584416	0
crates/cairo-lang-sierra-to-casm/src/invocations/range.rs	3388	ad97da4048ea0c37ba113aed1f6cc196439c334a6e67f549dcd610954be937e1	0
crates/cairo-lang-sierra-to-casm/src/invocations/range_reduction.rs	4930	d5fcf66b9e18fbe2437b0dd28ac51be757964a8d928a4f30a681244114cb3da7	0
crates/cairo-lang-sierra-to-casm/src/invocations/squashed_felt252_dict.rs	541	b0660d114e0d9feec38c8d4b8bda36f6c1069984de64451d1dce85722a21dd76	0
crates/cairo-lang-sierra-to-casm/src/invocations/starknet/mod.rs	8819	6777a75df5be190a86110e9a38d138aa0c668766dae38dd6c194ba6e4cba7556	0
crates/cairo-lang-sierra-to-casm/src/invocations/starknet/secp256.rs	1966	0bae520e54dfa5e86c20df4cd3ebe6feb96be67c71fd89b0ce43cfc4547a20f3	0
crates/cairo-lang-sierra-to-casm/src/invocations/starknet/storage.rs	3021	149a0623dc70d52ac813f420759ce7b8f72405f39622629ace2a22a064fce9d4	0
crates/cairo-lang-sierra-to-casm/src/invocations/structure.rs	4193	80a8d0fa76c6938e69a94874754eb629ac953e4c561580c9ca7dfb7f4748adb1	0
crates/cairo-lang-sierra-to-casm/src/invocations/trace.rs	876	765e19ebf445c4a3e70739ff101746435b186a6e10063c85f05661f879139982	0
crates/cairo-lang-sierra-to-casm/src/invocations/unsafe_panic.rs	562	fd644ddc10b083c4b7a556baa3119daed4014bb163b5935db6594f8400983d5b	0
crates/cairo-lang-sierra-to-casm/src/lib.rs	317	3a6d2b2e4cdb08b80214a495270ffc17f8eeda8a980a73105238a037f078976f	0
crates/cairo-lang-sierra-to-casm/src/metadata.rs	6255	72e7d5a09bbcddf2836fd0f720cd0688cf99bb59bc885e58fefe764c50639e31	0
crates/cairo-lang-sierra-to-casm/src/references.rs	7424	72acc4d75f259468939f68df9cc79307e69ea6ba59866fa69eaf3a38e184729e	0
crates/cairo-lang-sierra-to-casm/src/relocations.rs	6266	c0ff6f1b2ffc92e74328d662a1d454d85e43252b72cd30d344eea3ee8636d180	0
crates/cairo-lang-sierra-type-size/Cargo.toml	366	240984359ba8a1e4f73a9305df42751d97ac326c34f45f4e64b39d356ad56d16	0
crates/cairo-lang-sierra-type-size/src/lib.rs	7592	db78a7ef6c705b0d45ba782b576550ff092be1b1094b93458b0967d6f09b6ac4	0
crates/cairo-lang-starknet/Cargo.toml	2142	6a1360c1cca1097b573b751d708918ff71fa8092c1b17dadb8ec8cfdfbef7ea5	0
crates/cairo-lang-starknet/src/abi.rs	41779	381a37251032046c995613128b75696940e06343ed8415a25fc2aca474ea86ad	0
crates/cairo-lang-starknet/src/aliased.rs	649	38da360be02f8e593caaf6b3e2fd9c120040cddfa7e43bdb5a0f95d73289b937	0

crates/cairo-lang-starknet/src/analyzer.rs	13388	9a9d9a06288c9beba04cb98460e8884a9432272cfebdefeb2b96aab758f13422	0
crates/cairo-lang-starknet/src/compile.rs	11380	9933fdb854db7fd63bb9d0f20bee87757b9b75a83dc0650e57f3d7b49b9ffcf5	0
crates/cairo-lang-starknet/src/contract.rs	16046	c3cd61b1b1cbebe22d82ed0c02c77019d468448157f6f9d98f4b06af2a4d542b	0
crates/cairo-lang-starknet/src/inline_macros/get_dep_component.rs	4687	81a782b89f8cf97f6354eaeff56c814f5dd65c2f2dac2f4d79af6d7f94268a1a	0
crates/cairo-lang-starknet/src/inline_macros/mod.rs	45	16fd1d759add691ed9c24b26dda646ad9eda79fb7240f14c752ac74bdedc7637	0
crates/cairo-lang-starknet/src/inline_macros/selector.rs	2307	cca3bf5b0a97993fc98aecea0036e6bf767fb0d7cefc0a202b3feca161240e31	0
crates/cairo-lang-starknet/src/lib.rs	1182	c348847751678d4de26eebd99d5123b4c25c071f36de8fd435981e9d09790e45	0
crates/cairo-lang-starknet/src/plugin/aux_data.rs	1541	81291dd720309acabb0c0a26222be996df1853eb058c5654b0fb0d9e3ee555a7	0
crates/cairo-lang-starknet/src/plugin/consts.rs	3802	3eaae97df33b2bfa30bb96e6c0123ff3433ac797db3af9670f636cce5acd8098	0
crates/cairo-lang-starknet/src/plugin/derive/event.rs	14301	e8647c159b45677bb7e46110bf9c597a7fbba459be52698dc8c93c525071d731	0
crates/cairo-lang-starknet/src/plugin/derive/mod.rs	2050	6227361b246cf3c4524c080c65512dc6cb06b8550e6c09785c7497b516249622	0
crates/cairo-lang-starknet/src/plugin/derive/store.rs	15777	a7eff76b3d88daa8b417f7b2e860a862866cdd1cdc5851a151705593b26b149a	0
crates/cairo-lang-starknet/src/plugin/dispatcher.rs	27474	7fdda5a7c96461c76447035b6e211e65b4822429a20dec4f13e0ee81b86fd3d2	0
crates/cairo-lang-starknet/src/plugin/embeddable.rs	9055	4d51e3fc753f7b0777881054db9c4d5b07d53c3c03517f0b1fadb75f43e45dc6	0
crates/cairo-lang-starknet/src/plugin/entry_point.rs	15108	3fc15ee6c59e2a4b0a798dd25a56eafd9063df17a2d3e205dab21d5ead226dca	0
crates/cairo-lang-starknet/src/plugin/events.rs	3925	454e0ffa68531a86d1f889feb387db5b20760a90728469cc9c21fbb87ae00c44	0
crates/cairo-lang-starknet/src/plugin/mod.rs	3736	91f056fae747fc9cef893b8d868fcef55fe17aee51094fc8aa037ea38ad07a0e	0
crates/cairo-lang-starknet/src/plugin/starknet_module/component.rs	18599	c6a336d75e0571f1ba9325648a680602e954f7906541e829fab22f1e61ab14b3	0
crates/cairo-lang-starknet/src/plugin/starknet_module/contract.rs	32103	c61fc50f0f6d6ca2a1ee5cf84081a722a503fe837c064ffe1c0a285152f45499	0

crates/cairo-lang-starknet/src/plugin/starknet_module/generation_data.rs	2813	c53524f10caaf387d871216808591c295f37b3126e131626479873ddb064653e	0
crates/cairo-lang-starknet/src/plugin/starknet_module/mod.rs	9931	6617015634a6f58154d3becfec6fe381409de9b325a8e3e637117ed337138c34	0
crates/cairo-lang-starknet/src/plugin/storage.rs	9473	6e1536d861dd31cef2eaeb8af42cab2972afaef82538fac551434b9b146f62d6	0
crates/cairo-lang-starknet/src/plugin/storage_interfaces.rs	36520	762e142ff07dee6f49a887c2c7e12f03683af522b5c263485fe3528693a20291	0
crates/cairo-lang-starknet/src/plugin/utils.rs	8745	8a48739b3844d3f6466d5daf32b0337e93856427ee94b4ebf371d948fe914ec6	0
crates/cairo-lang-starknet-classes/Cargo.toml	1430	dbaa0518f6a005070eccfeef8e20dc4703f55edd1d7bc2afb52009038f883d19	0
crates/cairo-lang-starknet-classes/src/abi.rs	6334	239701f9af14e798d7652ce509adb8c9663b16de1aba4087f2b0232910834a90	0
crates/cairo-lang-starknet-classes/src/allowed_libfuncs.rs	5759	f04ae3fa8baaff30672e1fe397b9c6c764b9e646d73efea19bfe381a9e9afaac	0
crates/cairo-lang-starknet-classes/src/casm_contract_class.rs	27208	bbf1f0e9d0cff5d47002a5efbea90b42d0933cc24a742895777423f9f2a53532	0
crates/cairo-lang-starknet-classes/src/compiler_version.rs	1496	47ed2a5b52f0e682d83edccdd06f78cda603fa2f754f16285739fd42c7cd4a74	0
crates/cairo-lang-starknet-classes/src/contract_class.rs	6618	126d7933e29c75ee15e9887cd6fe1904e2a7e5fbf38e624c105fcdb0fc3e3da1	0
crates/cairo-lang-starknet-classes/src/contract_segmentation.rs	7240	8fed6d566fc8b705c938cdf113af1f026f1c13d6b4db6fcca3121552550e4abf	0
crates/cairo-lang-starknet-classes/src/felt252_serde.rs	24171	4d25743988ab34f0859ca27de6c9cabbad4a446cd8be244119e0e1bf9dc65a9b	0
crates/cairo-lang-starknet-classes/src/felt252_vec_compression.rs	3910	7d5de620428ea451b8d3097003e25594c9db2a5edb6d7aab8369db8826d85c67	0
crates/cairo-lang-starknet-classes/src/keccak.rs	445	5c1f594e59fc02b0ed542f735856072afb4f72f139faa6b81fce5f295a5e6f84	0
crates/cairo-lang-starknet-classes/src/lib.rs	570	b521745e338003fb09016ab7de0cca3c78154897741907c2f751a6c5c187bc7f	0
crates/cairo-lang-syntax/Cargo.toml	1035	eda9a67c2b8c7ca4134d9c76b36546df594a16886a8302ac79dcc2c7f2f2d556	0
crates/cairo-lang-syntax/src/attribute/consts.rs	3199	ecc03a3bd97576d00dcf18379d29f6239ccda22ea79fd24c42798f43d82aa48f	0
crates/cairo-lang-syntax/src/attribute/mod.rs	36	6616808eb02f9517ce92123f6af2c82fb04de0badf90321190bc7961e22f9943	0

crates/cairo-lang-syntax/src/attribute/structured.rs	6238	b4159474031392ebf99a707cf828753c9b7a0a9cd58f70df4cf675405e189ab1	0
crates/cairo-lang-syntax/src/lib.rs	108	43b235fe8a4e879d568641754be85aca2fd01cfa85150dcc7a490e22c4432f4b	0
crates/cairo-lang-syntax/src/node/ast.rs	1587267	b26a1df700db9d3b96fb5a1ee10a1a16c174858a0807d4ff40bdc2d622a67234	0
crates/cairo-lang-syntax/src/node/ast_ext.rs	4359	3cf597a94388635e92cbb989a6422c59baec18ee1f5e9ea143251b22c5482fd3	0
crates/cairo-lang-syntax/src/node/db.rs	327	996b0488dd22cd764af7e68663e71d27795d8df3391ad1ffc3c1da68cacda1f2	0
crates/cairo-lang-syntax/src/node/element_list.rs	1800	14c40b5bcb89c7fa680ab2b6e893914d2b45c3b4dc299e7a81cdc108d440f61f	0
crates/cairo-lang-syntax/src/node/green.rs	1114	93d75c996ecfbce693e8e43229d4dc5afd9cc9bfe790693f4234df8de0fe0dc2	0
crates/cairo-lang-syntax/src/node/helpers.rs	44261	16994ab0212961a8cd6a4d8da6cdced7d9803aab3c5b45a0265a6a55153d1a91	0
crates/cairo-lang-syntax/src/node/ids.rs	2186	3689e50a47fbc539c418ff56e5e53b35df62ec324c410e5100c704fa0def6833	0
crates/cairo-lang-syntax/src/node/iter.rs	3789	76cf8662e61ad175f84201894f84b951fd79a4f49fae4070a2e8fdfa315ff1fe	0
crates/cairo-lang-syntax/src/node/key_fields.rs	16798	bf6d4796017cc958799a7ccc384356124a0a51abe0e892dc8073a40ed49fb2af	0
crates/cairo-lang-syntax/src/node/kind.rs	18270	aaf0845d07a0979fdb67be477255d41b1580b30b064d217baa4d8472ef48c661	0
crates/cairo-lang-syntax/src/node/mod.rs	36780	5397d59a3439cf29c5540c0d840c4785401e179a87aa30f3ffa541a872d3c766	0
crates/cairo-lang-syntax/src/node/stable_ptr.rs	962	c5bd56f99e85bcd2b0a503a1aaa82b2428a09543fea1ee722e927c13e755b3a9	0
crates/cairo-lang-syntax/src/node/with_db.rs	4853	3a56dca38c04674b5dfed9d43f512c9a907b25732da969313d5e255f3db33b7b	0
crates/cairo-lang-syntax-codegen/Cargo.toml	560	6465f50eb26d90b340b11173ddc19772a701df2c7e32dccdd43c33e2173f9f7d	0
crates/cairo-lang-syntax-codegen/src/cairo_spec.rs	51368	d679af4a2baf50c791a1d34c71935f63367c1f0cb059b2d5e6af4038b59900d8	0
crates/cairo-lang-syntax-codegen/src/generator.rs	31277	14a8766462d4ed9152b796a4cca8dcd5241b3128201ddd931814fd8db43b23aa	0
crates/cairo-lang-syntax-codegen/src/lib.rs	128	d8aa0047699da6e84a71b30f660d2f7333793f8d86250f1a93d3bf238f990ca2	0
crates/cairo-lang-syntax-codegen/src/spec.rs	7992	3338b1ad4c56675835596855e8eb055754652ec3b64a586b08d1e41751f769c0	0
crates/cairo-lang-test-plugin/Cargo.toml	1581	b1056e8fb5df40d95dbc246d3bb01613699f18993930fb4645718063a6296f15	0

crates/cairo-lang-test-plugin/src/inline_macros/assert.rs	10538	6e130200a84abf3ded29ccf89ee9a1783a3c0eb19b7d88a14af894b8b0dc9deb	0
crates/cairo-lang-test-plugin/src/inline_macros/mod.rs	16	f8d0c219a61c7f7865511f7b329744702d7cc68aaf1200449a0a86770386b4a7	0
crates/cairo-lang-test-plugin/src/lib.rs	12890	feba47115073e710f534246a8a24b6bc0b7d763fd90123f284299e37f3e2ee36	0
crates/cairo-lang-test-plugin/src/plugin.rs	1416	6a33d61d518943b7c36ad383dee8033219065edd9472a02ef8924ee43249269e	0
crates/cairo-lang-test-runner/Cargo.toml	1248	4d18dc8ff3511bff3818cea60b3832a0660095e6351518a99203a2eb41d59ebd	0
crates/cairo-lang-test-runner/src/lib.rs	19871	1c9cff406327f41db978a5671fa62f96380bc672c01e977c00c615f9306b4287	0
crates/cairo-lang-test-utils/Cargo.toml	1325	c3eb4b1d2ed4161cf9f3e06aed0e421921c876a2ac9bda8e10c9828966f8f152	0
crates/cairo-lang-test-utils/src/lib.rs	4970	459dbf11d40b74b139ef405df52e0df64d14cf67380224eb9cd36601ab4f9918	0
crates/cairo-lang-test-utils/src/parse_test_file.rs	17539	b3feee64aaa5083eed39d204000f3c4672f5966375e7352e19d2ba4b0bb57369	0
crates/cairo-lang-utils/Cargo.toml	1515	91662f18231120a839ea0b744f93bc4e79a5a4c28747b714fb2974f3705fed13	0
crates/cairo-lang-utils/src/bigint.rs	7681	467462418564a70118ddc9c606c09e9affc7ad32f47cda10dfe50660befadbe6	0
crates/cairo-lang-utils/src/byte_array.rs	197	2fea15a510ae3b46b4391ce7fd1da8dce1610112af8d035ce28a6662da4a80f6	0
crates/cairo-lang-utils/src/casts.rs	622	b2948700af813fd93db652fde0c5034059d0120103c7d212511646ec18629947	0
crates/cairo-lang-utils/src/collection_arithmetics.rs	4302	64b733f781074df0ef3b02fd311dd43741bccb30b850a7ed77625c79e7b61e82	0
crates/cairo-lang-utils/src/deque.rs	1794	2f186c8c526e45c1ab4800054ed09db02f426be213bbeed45a587d801756370e	0
crates/cairo-lang-utils/src/extract_matches.rs	1904	7d4d4287530787ae8fcc79f1b527a45e53fbc9870a77cb6b2fc30a67828a4510	0
crates/cairo-lang-utils/src/graph_algos/feedback_set.rs	4244	084b1a4c871046016b2d02296b7314da9855bb0a9fb9de6f3992a0614ed4c72f	0
crates/cairo-lang-utils/src/graph_algos/graph_node.rs	1219	84b31e1b638c821fd8b71c1b3fbb756a6245dcfb94e4c1636915fa611b3bcf32	0
crates/cairo-lang-utils/src/graph_algos/mod.rs	105	484aef560b14e85e56dbd6e4c46d4472cacd651c8a992ab9f7960552f40d6f01	0
crates/cairo-lang-utils/src/graph_algos/scc_graph_node.rs	849	b984f5b04f391a21fe3b560b30c6cff3616e35ee1fb3ff73bcc2ab8c9614ffd6	0
crates/cairo-lang-utils/src/graph_algos/strongly_connected_components.rs	4859	79e4e20fa47056c42537109a2480e78e300fc11ad73170c1470ab749edab606d	0

crates/cairo-lang-utils/src/heap_size.rs	10993	ed70e37d8e9b8a2c4980ae81204ec8edd638adc30b33a39866a89b1ea582024c	0
crates/cairo-lang-utils/src/iterators.rs	313	c451ca43774c3cb5df7eca24e63c8c9a7db96b955698f6b070ba28d277ce8883	0
crates/cairo-lang-utils/src/lib.rs	5081	903884ce4c523e5bca7b6f420a7f8bde0d82656665353f98d81f12c2925c29de	0
crates/cairo-lang-utils/src/logging.rs	2707	a485c2cd4cdb89964382a738d88923db53ebd6ce64cbdf12c398c0248d048295	0
crates/cairo-lang-utils/src/ordered_hash_map.rs	4850	80efafe352cc58037414acd1cfa8d0a1deb530e81d19a743ce90c9cf031de5c7	0
crates/cairo-lang-utils/src/ordered_hash_set.rs	3010	3d92b201071e5ffe2b5643fb24fb79660111db41814caaefc94cb185b7376922	0
crates/cairo-lang-utils/src/small_ordered_map.rs	2426	5e4383edcfe26ba4b9efa5ed60fc75492d210a4ec96f7c6ff21d8d8083c57ea6	0
crates/cairo-lang-utils/src/unordered_hash_map.rs	12795	5465e8f07fe63cad2291180063ebe136616753637689d4f3c7d6d109ffe924bc	0
crates/cairo-lang-utils/src/unordered_hash_set.rs	3763	68cb657c870203a100879692ed73b5fe90c36238bc1f38e32b096c7fbdfe60e3	0
ensure-no_std/.cargo/config.toml	42	285033faa15221410d801d5394381ae495890d37cc60b6c0aff0b7d08beda7b8	0
ensure-no_std/Cargo.toml	391	b476920fbefde741529a43a0f9fa7da09e070ff01b5385b9a625133c5f2d0bb0	0
ensure-no_std/src/main.rs	447	8cc2e4ea686f58cf1ae1db7029276784d5b9f107cb65197959896ca2d08ea1df	0
scripts/check_comment_punctuation.py	5854	57f582217517901ec58426679b422ac4261287ef27dbda3d058d5e77d37ce7c8	0